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The role of generative AI in R&D sensemaking

An exploratory study on individual and collective sensemaking practices under uncertainty

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Abstract

Research and Development (R&D) work is often characterized by uncertainty, requiring organizational actors to interpret ambiguous or incomplete information before acting on it. Even though AI is increasingly used in R&D, research on how it enters sensemaking is limited. This thesis therefore examines how the use of generative AI tools shapes practices of individual and collective sensemaking in R&D. The study adopted an exploratory empirical design based on two industrial firms with operations in Sweden. Data from 25 semi-structured interviews with 18 individuals was analyzed abductively, combining a literature informed framework with the themes emerging from the empirical data. The findings suggest that generative AI tools were primarily used for exploratory and interpretive sensemaking activities, such as information search, orientation in new domains or generation of new perspectives. Its role in collective sensemaking appeared more indirect, as generative AI primarily supported individual preparation for team discussions rather than entering them as an autonomous authority. Confidence in AI-supported sensemaking depended on contextual fit, the possibility of verifying outputs, the user's domain expertise and the stakes of the task. Generative AI did therefore not remove uncertainty but rather shifted it from limited information access toward concerns about the reliability and contextual appropriateness of AI outputs. The study nuances research on AI in R&D sensemaking by indicating that generative AI involvement is conditional across sensemaking stages. Further, based on the findings a framework is proposed relating the roles of generative AI in R&D sensemaking to contextual fit and accountability demand. The study suggests organizations can enhance the value of generative AI by providing secure access to relevant organizational knowledge and strengthening employees' capabilities to critically evaluate AI outputs.

Keywords

Generative AI; Research and development; Sensemaking; Uncertainty; Qualitative study

Sammanfattning

Forsknings- och utvecklingsarbete (FoU) kännetecknas ofta av osäkerhet, vilket kräver att organisatoriska aktörer tolkar tvetydig eller ofullständig information innan de agerar utifrån den. Även om AI används i allt större utsträckning inom FoU är forskningen om hur den bidrar till meningsskapande begränsad. Denna avhandling undersöker därför hur användningen av generativa AI-verktyg formar praktiker för individuellt och kollektivt meningsskapande inom FoU. Studien använde en explorativ empirisk design baserad på två industriföretag med verksamhet i Sverige. Data från 25 semistrukturerade intervjuer med 18 individer analyserades abduktivt, vilket kombinerade ett litteraturinformerat ramverk med de teman som framkom från den empiriska datan. Resultaten tyder på att generativa AI-verktyg främst användes för explorativa och tolkande meningsskapande aktiviteter, såsom informationssökning, orientering i nya domäner eller generering av nya perspektiv. Dess roll i kollektivt meningsskapande verkade mer indirekt, eftersom generativ AI främst stödde individuell förberedelse för teamdiskussioner snarare än att ingå i dem som en autonom auktoritet. Förtroendet för AI-stödd meningsskapande berodde på kontextuell anpassning, möjligheten att verifiera output, användarens domänexpertis och uppgiftens insatser. Generativ AI tog därför inte bort osäkerheten utan flyttade den snarare från begränsad informationstillgång till oro för tillförlitligheten och kontextuell lämplighet hos AI-resultat. Studien nyanserar forskning om AI inom FoU-meningskapande genom att indikera att generativ AI-involvering är villkorad över olika meningsskapande stadier. Baserat på resultaten föreslås vidare ett ramverk som relaterar den generativa AI:s roller i FoU-meningskapande till kontextuell anpassning och ansvarsskyldighetskrav. Studien föreslår att organisationer kan öka värdet av generativ AI genom att tillhandahålla säker tillgång till relevant organisatorisk kunskap och stärka medarbetarnas förmåga att kritiskt utvärdera AI-resultat.

Nyckelord

Generativ AI; Forskning och utveckling; Meningsskapande; Osäkerhet; Kvalitativ studie

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1 Introduction

Artificial Intelligence (AI) could arguably bring about substantial changes and potential in various areas of private and business life. In knowledge intensive work, such as Research and Development (R&D), recent research argues that use of AI tools in organizations may improve new product development process and R&D efficiency (Cooper, 2024a; Liu & Li, 2025) and is positively associated with innovation output (Liu & Li, 2025; Rammer et al., 2022). AI adoption at a sufficiently high intensity was shown to often relate to better financial outcomes (Lee et al., 2022). Dimensions such as innovation and efficiency are important for R&D performance. However, R&D also often is strongly affected by uncertainty (Bloom, 2007). We must therefore understand how organizational actors transform uncertainty into actionable interpretations. This process is referred to as sensemaking (Weick et al., 2005). Ever since the concept of organizational sensemaking was established (Weick, 1995), an extensive body of literature has developed around triggers, processes, roles and practices (Maitlis & Christianson, 2014). Yet we know little about how the use of contemporary generative AI tools influences sensemaking practices. This thesis addresses this gap. Through a qualitative study in two Swedish industrial organizations, this thesis examines the effect of generative AI tool use on different sensemaking activities in R&D.

1.1 Background and Problem

R&D is often strongly affected by uncertainty (Bloom, 2007), requiring development teams to act despite incomplete information (MacCormack & Verganti, 2003). In such settings, the challenge is often not only to reduce uncertainty but also to acknowledge it (Lipshitz & Strauss, 1997), and to use available information for constructing a shared understanding of what is relevant and plausible (Weick, 1995). This is conceptualized as the practice of sensemaking, describing how actors notice cues, negotiate plausible interpretations and act on these in an ongoing cycle (Maitlis & Christianson, 2014; Weick et al., 2005).

At the same time, AI is increasingly examined in R&D-related contexts (Cooper, 2024a; Doron et al., 2025) and discussed in terms of adoption, capabilities and outcomes (Gama & Magistretti, 2025; Lee et al., 2022). While these perspectives are highly relevant, AI is also likely to reshape the collective sensemaking processes that enable coordinated action under uncertainty. First, AI may help overcoming human information processing constraints (Haefner et al., 2021), and potentially contribute to richer interpretations. At the same time, algorithms can also influence what information is perceived as relevant, thereby steering what drives the discourse (Gillespie, 2013). Second, AI output becomes an object of interpretation itself, particularly when users must assess its reliability and relevance under uncertainty (Joussen et al., 2025). Third, AI may change dynamics through which teams converge on a shared interpretations that lead to decisions (Ben-Shahar et al., 2025; Suri, 2025). Finally, decisions must be perceived as legitimate to relevant stakeholders (Ruef & Scott, 1998). Organizational legitimacy depends on a perception of appropriate alignment with standing norms and beliefs (Suchman, 1995) and, for algorithm-augmented decision making, is moderated by transparency and explainability (de Fine Licht & de Fine Licht, 2020) as well as reviewability and decision importance (Martin & Waldman, 2023). This adjacent research provides insights into AI-supported mechanisms relevant to sensemaking, but the insights remain fragmented and do not view AI explicitly as part of sensemaking in R&D. This leaves limited understanding of how AI enters day-to-day sensemaking practices and how it affects different sensemaking activities under uncertainty. This thesis addresses this gap by examining empirically how generative AI tools are used around sensemaking in R&D, and in particular how generative AI use influences exploration, interpretation, team discussion and decision confidence.

1.2 Research aim

The aim of this degree project is to investigate how generative AI tools shape sensemaking-related work in R&D. The study examines how generative AI is utilized in practice across activities related to sensemaking. This includes exploration and information search, interpretation and articulation, preparation for team discussion and confidence and verification in AI-supported sensemaking. This aim was addressed through an exploratory study on two industrial firms with operations in Sweden, using semi-structured interviews and an abductive analysis.

This degree project contributes to both theory and practice of organizational management. The main contribution is the development of a conceptual framework for how generative AI reshapes sensemaking work in R&D. More specifically, the framework explains how generative AI enters sensemaking under different conditions of contextual fit and accountability demand. The framework was developed abductively starting with a literature informed draft to then develop it in accordance with empirical findings. The study further concludes practical implications for R&D practitioners.

1.3 Structure of the thesis

Chapter 2 presents relevant background information about uncertainty and sensemaking in R&D, as well as a literature synthesis including derived research questions and a conceptual framework. Chapter 3 features the methodology used to conduct research by addressing data collection, analysis and limitations besides a discussion of alternative approaches. Chapter 4 summarizes the empirical data and highlights key findings. In Chapter 5, the results are contextualized within the broader sensemaking literature, discussing the role of generative AI and relevant conditions. Chapter 6 assembles the empirical findings into a proposed refined conceptual framework. Finally, Chapter 7 features the conclusions, implications for theory and practice as well as limitations and future work.

2 Background

The background chapter is structured to build the analytical foundation of this study. It starts with a conceptualization of uncertainty in R&D as the condition that triggers the need for sensemaking. It then examines the theory of organizational sensemaking to establish a process for how actors engage in sensemaking. Building on these concepts, literature is discussed on how AI tools are used in R&D to identify where they may enter sensemaking activities as sources of information, interpretation support or decision aid. Literature on legitimacy and trust in AI is introduced to explain how the usefulness of AI tools does not automatically result in them being perceived as socially accepted input for decision-making. Finally, findings are synthesized into four resulting research questions that set a scope for this thesis.

2.1 R&D and uncertainty

Uncertainty is a fundamental concept in organizational theory. Research has shown that perceived uncertainty levels in an organization are associated with key firm activities, such as resource commitment decisions (Chen et al., 2024) or innovation outcomes in the form of patent filings (Yu et al., 2021). Uncertainty is therefore not just an abstract phenomenon, but a rather consequential issue organizations must deal with. Bloom (2007) argues that uncertainty often reduces responsiveness in R&D and can delay activities to adjust to changing conditions (Bloom, 2007). Because of the systematic effect on decision timing and outcome for organizations, it is necessary to clarify what uncertainty means and how it appears in R&D.

Types and sources of uncertainty

Milliken defines uncertainty as *"an individual's perceived inability to predict something accurately"* (Milliken, 1987). He stresses that uncertainty is more a perception dependent on the individual lack of information and the inability to distinguish between relevant and irrelevant information, rather than only a general condition (Milliken, 1987). Building on this definition, we must acknowledge that uncertainty is not a singular concept but takes different forms that matter for managerial action. Milliken introduces three types of uncertainty about the environment: State uncertainty affects the ability to predict components of an organizations' environment, effect uncertainty affects the ability to accurately predict the impact of particular events on an organizations, and response uncertainty describes a lack of knowledge about an organizations options to respond to an event (Milliken, 1987). Milliken's classification distinguishes forms of uncertainty, but we must also consider where uncertainty originates. While early literature places a major focus on the environment, Priem et al. (2002) identify both internal and external sources of perceived uncertainty among executives (Priem et al., 2002). They show that uncertainty can emerge through industry players or regulatory developments, but also from internal factors, for example in human resource management or when estimating production costs (Priem et al., 2002). Other research investigates the underlying coping mechanisms. Drawing from qualitative data of 102 retrospective self-reports, Lipshitz & Strauss (1997) find that decision-makers deal with uncertainty by either trying to reduce it, by acknowledging and dealing with it or by suppressing it (Lipshitz & Strauss, 1997).

Uncertainty in R&D

These concepts become especially significant in R&D, where uncertainty is structural and consequential. In an industrial context, Xing et al. (2020) characterize traditional development processes with long cycle times, low success rates and difficult management needs (Xing et al., 2020). Higher uncertainty levels may reduce a firms responsiveness toward changing conditions (Bloom, 2007) and push firms to make changes to product design at later stages of development (MacCormack & Verganti, 2003). Research suggests that development project teams should respond to uncertainty

sources proactively in early project stages to maintain flexibility in the development process (MacCormack & Verganti, 2003).

Conceptualization of uncertainty used in this thesis

The diversity of lenses and definitions, through which uncertainty is observed and described in literature, raises the need to derive a conceptualization of uncertainty for this thesis. Following Milliken's (1987) perceptual view, uncertainty is conceptualized broadly including missing information and equivocal or ambiguous interpretations. This use should be distinguished from conceptualizations that separate uncertainty and ambiguity more explicitly. Lipshitz and Strauss (1997) propose a conceptualization of uncertainty as a *"sense of doubt that blocks or delays action"* (Lipshitz & Strauss, 1997). This includes three important implications: Uncertainty is subjective, it can stem from various sources of doubt and it is consequential because it constrains action (Lipshitz & Strauss, 1997). Combining these concepts, I derive a conceptualization of uncertainty:

> *Uncertainty is a subjective sense of doubt, arising from missing information or equivocal interpretations, that blocks or delays action by limiting understanding of environmental states, their implications or response options*. (Lipshitz & Strauss, 1997; Milliken, 1987)

The literature above conceptualizes different types of uncertainty as perception-based and shows underlying coping mechanisms. However, coping with uncertainty in R&D teams is rarely done individually. Teams must jointly interpret and negotiate cues and build a plausible story to justify actions. To explain this collective interpretation process, the next chapter deals with organizational sensemaking (Weick, 1995; Weick et al., 2005).

2.2 Sensemaking in R&D

Rather than rational decision models, the concept of sensemaking explains how actors deal with uncertainty, ambiguity and incomplete information (Maitlis, 2005; Maitlis & Christianson, 2014; Weick et al., 2005). Sensemaking describes how organizational actors turn environmental circumstances into meaning that enables coordinated action in ambiguous situations (Weick et al., 2005).

The concept of sensemaking under uncertainty

Drawing from Weick et al. (2005) this thesis distinguishes three closely related sensemaking activities. First, noticing and bracketing involves extracting cues from the environment, (2) interpreting is a process of selecting a plausible narrative that explains the situation and allows to (3) take action on it. Action feeds back into the process by changing the observed environment, resulting in an ongoing, recursive process (Weick et al., 2005). The sensemaking process is triggered when expectations are violated or upon occurrence of significant, yet ambiguous events, often involving a threat to existing identities and understandings (Maitlis & Christianson, 2014). This underlines the central role of sensemaking in highly uncertain environments like R&D.

A central contribution of Weick (1995) is the specification of sensemaking characteristics. He characterizes the process as *"(1) grounded in identity construction, (2) retrospective, (3) enactive of sensible environments, (4) social, (5) ongoing, (6) focused on and by extracted cues [and] (7) driven by plausibility rather than accuracy"* (Weick, 1995). This implies that sensemaking is perceptual. Actors selectively extract cues and interpret them based on their own identity perception and experiences, which leads to different individuals noticing and bracketing different aspects of the same situation (Weick, 1995). To enable collective action, it is substantial that shared meaning is created that accommodates for these subjective perceptions, which is why plausibility is often favored over accuracy (Weick, 1995). Through arguing, influencing others, forming expectations and committing

to an action, actors converge to what Weick calls *"generic subjectivity",* supporting coordinated action under ambiguity (Weick, 1995).

Collective sensemaking and sensegiving

Because sensemaking is social and requires acceptance of different actors, meaning does not emerge automatically from individual interpretations but often has to be negotiated (Maitlis & Christianson, 2014). Meaning is therefore actively shaped, as some actors attempt to influence how others interpret cues (Gioia & Chittipeddi, 1991; Maitlis, 2005). This introduces the closely related concept of sensegiving, which highlights how actors frame events and guide collective interpretations to enable certain desired actions (Gioia & Chittipeddi, 1991). Gioia and Chittipeddi (1991) describe a process, in which leaders first make sense of events and then communicate their interpretation to stakeholders, who then themselves interpret and respond, trying to influence the collective interpretation (Gioia & Chittipeddi, 1991). Rouleau (2005) found, that managers even adjust their narrative and emphasis on specific cues depending on the audience they are trying to influence (Rouleau, 2005). Importantly, sensegiving is not limited to senior leaders, as in practice middle managers often act as mediators who translate interpretations across groups (Rouleau, 2005). Building on this, Maitlis (2005) introduces different forms of sensemaking/sensegiving dynamics dependent on the involvement of leadership and stakeholders (Maitlis, 2005). While high levels of leadership sensegiving typically result in controlled processes and high interpretation convergence, high stakeholder sensegiving accounts for the integration of more individual views (Maitlis, 2005). In practice, sensemaking often takes the form of either high leadership sensegiving with limited stakeholder sensegiving, producing unitary yet narrow accounts, or limited leadership with strong stakeholder sensegiving, resulting in multiple, fragmented accounts and less consistent actions (Maitlis, 2005).

Language and framing for interpretative alignment

A key tool in the process of sensemaking is language. When issues are novel or abstract, actors make unfamiliar situations tangible, for example by constructing mental models and communicating them in metaphors (Hill & Levenhagen, 1995). This framing supports alignment among actors as it turns abstract issues into a medium more easily interpreted by humans (Hill & Levenhagen, 1995). At the same time, different actors may introduce different frames for the same situation, which implies that multiple plausible narratives can coexist (Maitlis & Christianson, 2014; Weick, 1995). Cognitive alignment is therefore not default but may require negotiation and ongoing adjustment (Maitlis & Christianson, 2014). Friis-Holm Egfjord and Sund (2020) highlight the importance of aligned perceptions, showing how misaligned perceptions of environmental change between an innovation department and core business unit of an insurance organization hindered the pursuit of radical innovations (Friis-Holm Egfjord & Sund, 2020).

In summary, sensemaking research shows how actions taken by teams and organizations under uncertainty depend on how cues are selected and interpreted in given situations. With the increasing use of AI tools in R&D work, they are likely to affect sensemaking as well as perceptions of plausibility and uncertainty. The next chapter therefore investigates AI use in R&D regarding uncertainty and organizational sensemaking.

2.3 Use of AI tools in R&D

First, there is a need to create an understanding about what I refer to as AI tools during this thesis. A detailed technical elaboration on types or definitions of artificial intelligence lies beyond the scope of this study, but instead the thesis uses a functional working definition derived from Kalota (2024). Artificial intelligence can be used as a broad term for computational systems that perform tasks normally associated with human cognitive activity, such as pattern recognition, interpretation and

content generation (Kalota, 2024). While the background chapter draws on broader literature on AI in R&D, the empirical focus of this thesis is generative AI. Accordingly, references to AI tools in the empirical chapters primarily address generative AI tools, unless broader AI systems are explicitly specified. Generative AI refers to a form of Machine Learning, which itself is a subset of artificial intelligence, that is often built on artificial neural networks or large language models (LLM) and is capable of generating new outputs in response to user prompts, including text, code or designs (Kalota, 2024). This working definition also takes into account both the roles of AI as an originator, expanding search spaces and ideation and AI as a facilitator, improving efficiency and supporting optimization of processes (Brem et al., 2023). It further includes Berente et al.'s (2021) understanding of the unique characteristics of contemporary AI, namely autonomy, learning and inscrutability (Berente et al., 2021).

Existing literature suggests, that AI may support a range of innovation- and R&D related activities, including how firms search and evaluate ideas, carry out experimentation or develop new products (Brem et al., 2023; Cooper, 2024a; Doron et al., 2025; Haefner et al., 2021). Empirical research suggests that investments in AI and enabling complementarities are often positively associated with financial outcomes (Lee et al., 2022). Other research links AI use to higher innovation output (Liu & Li, 2025; Rammer et al., 2022), increased product development process and R&D efficiency (Cooper, 2024a; Liu & Li, 2025) and reduced costs related to AI-enabled process innovation (Rammer et al., 2022).

As partially referenced above, a large body of literature indicates a considerable effect of AI on R&D practices, especially regarding technology adoption, capabilities and innovation outcomes. This chapter first summarizes relevant research on AI use in R&D. Building on this, it links AI use to uncertainty and organizational sensemaking to investigate how AI use changes the collective processes through which R&D teams handle uncertainty and justify decisions.

2.3.1 AI use throughout the R&D process

Literature suggests a broad range of R&D activities that benefit from the use of AI tools. Truong and Papagiannidis (2022) review literature on how AI may enable activities along different stages of the Innovation process. They indicate that AI can (1) support discovery by analyzing large unstructured datasets, (2) augment idea screening, (3) select promising test setups and (4) improve product development and commercialization processes (Truong & Papagiannidis, 2022). Even though the work focuses on innovation rather than R&D processes, the stages are still highly relevant for use in this thesis as there is a considerable overlap of important activities between those two adjacent domains. Moving forward in this chapter I will therefore take over the four segmentation steps used by Truong & Papagiannidis (2022) to assess literature around AI use cases

AI in discovery and idea generation

In the discovery and idea generation stage, AI may widen search spaces and help process large amounts of unstructured data (Truong & Papagiannidis, 2022). This aligns with work on innovation analytics, which highlights how AI can analyze large datasets to help practitioners identifying opportunities that would otherwise be costly to extract manually (Kakatkar et al., 2020). Broekhuizen et al. (2023) highlight how AI may help address human weaknesses such as information processing constraints but also biases when selecting partners for collaboration projects in an open innovation context (Broekhuizen et al., 2023). Other work indicates, that more targeted models can support managers to identify trends in emerging technologies (Mühlroth & Grottke, 2022).

AI in idea screening and selection

In idea screening and selection, AI may augment assessment and prioritization processes to help narrow choices before committing resources, by processing more data and thereby helping managers

make more informed decisions (Truong & Papagiannidis, 2022). Füller et al. (2022) argue that, while human expertise remains critical in idea selection, AI can augment this process by helping to overcome constraints caused by cognitive biases and limited information (Füller et al., 2022). From a theoretical perspective, Grilli and Pedota (2024) propose a potential of AI for augmenting convergent thinking in idea selection by finding patterns with historical ideas and estimating chances of success (Grilli & Pedota, 2024).

AI in experimentation

For experimentation, AI can support the selection of promising test cases and help accelerate experimentation cycles (Truong & Papagiannidis, 2022). One example is the discovery of potential drugs, where in practice AI is used to identify promising candidate substances for testing (Cooper, 2024a). AI can also support experimentation itself through the analysis of collected user data, for example on applications or websites (Verganti et al., 2020). Other research on drug development suggests that *"closing the loop"* of hypothesizing and testing can increase product development speed (Doron et al., 2025). In industrial product-development, AI-supported digital twins and simulation have been proposed to reduce the need for physical testing, thereby shortening R&D cycle times (Xing et al., 2020).

AI in product development

In later-stage development, AI may support product development decisions (Truong & Papagiannidis, 2022) and improvements to existing innovation processes product-development activities (Brem et al., 2023). Conceptual research in new product development suggests that AI has the potential to substantially decrease development times (Cooper, 2024a). Empirical work backs this, suggesting that AI-enabled process innovation was associated with a share of German firms' annual cost savings (Rammer et al., 2022). Liu and Li (2025) found that the intensity of AI use in Chinese firms increases the number of patents filed. In more competitive industries, AI application is thereby associated with more exploitative innovation, while in less competitive industries, it is associated with both explorative and exploitative innovation (Liu & Li, 2025).

Across all stages, literature suggests that AI can play a role as an originator expanding and generating solution spaces, as well as a facilitator improving existing processes and designs (Brem et al., 2023). In a broader sense, the originator/facilitator roles align with Raisch and Krakowski's (2021) augmentation-automation paradox. They argue, that AI may automate selected tasks, while simultaneously increasing the importance of human-machine interaction in adjacent consequential tasks (Raisch & Krakowski, 2021). Research in innovation management therefore emphasizes the need for AI-related enabling capabilities (Gama & Magistretti, 2025). Empirical work further identifies the need to build an understanding of specific AI implementation requirements (Füller et al., 2022), while conceptual work suggests broader organizational needs around data, technology and talent (Tekic & Füller, 2023). Finally, literature warns that AI projects are not bound to succeed, but can fail due to poor practices and insufficient implementation (Cooper, 2024b).

2.3.2 AI effects on uncertainty and sensemaking practices

The literature discussed in the previous section primarily focused on AI adoption, capabilities and outcomes. However, work on how AI influences the perception of uncertainty and the way individuals and teams in organizations make sense of it remains scarce. In this chapter I aim to narrow the lens and discuss what has been found about the impact of AI on the discussed concepts of uncertainty and organizational sensemaking, as introduced in sections 2.1 and 2.2 respectively.

The effect of AI on uncertainty

Wu & Shang (2020) conduct a literature study on AI-enabled decision-making and discuss the impact of AI on different types of uncertainty. Their findings suggest that AI can help managing uncertainty caused by a lack of information through its capability to process large amounts of data (Wu & Shang, 2020). However, they do also argue that AI introduces new risks, as it might fail to account for the diversity of human needs or struggle in unfamiliar situations encountered in the environment (Wu & Shang, 2020). This aligns with findings from Joussen et al. (2025). While the information processing capabilities of AI can reduce uncertainty and enhance transparency for highly complex problems, study respondents reported that AI's susceptibility to errors in low-complexity situations may lead to a perception of increased uncertainty (Joussen et al., 2025). Psychological studies show that people may lose trust in algorithmic forecasts more quickly than in human forecasts after observing errors, even if the algorithm proves to outperform the human statistically (Dietvorst et al., 2015). Because uncertainty is partly perceptual and interpretative (Milliken, 1987), these findings taken together suggest that AI may change what teams perceive as uncertain. Following this, the way in which R&D teams make sense of the changed uncertainty landscape may be affected as well.

AI in sensemaking: Cue extraction

The first key activity of organizational sensemaking is the extraction of cues from the environment (Weick et al., 2005). AI may help address human information-processing constraints by processing large amounts of data more efficiently than humans (Haefner et al., 2021). From a sensemaking perspective, this may allow AI to support cue extraction by expanding the environmental search space. Research on public relevance algorithms suggests that algorithms shape what information is treated as relevant for a given object of discourse (Gillespie, 2013). In an R&D sensemaking context, this may shape which cues are being noticed and brought into interpretation by the user. This perspective implies that AI can support uncertainty reduction by expanding environmental scanning, but it also steers attention and potentially narrows what is noticed.

AI in sensemaking: Interpretation

Once cues are extracted, one needs to interpret them to create a plausible base for action (Weick et al., 2005). Berente et al. (2021) argue that AI increasingly informs or automates managerial decisions (Berente et al., 2021). Applied to sensemaking, this suggests a need to interpret AI outputs themselves by reflecting on the assumptions they make and how they relate to the decision's context. Literature further highlights the role of a human interpreter. Hao et al. (2025) indicate that limitations of AI in causal reasoning, interpretation and alignment with norms require human judgement, especially regarding edge cases and consequential situations (Hao et al., 2025). Lay decision-makers have been found to engage in more analytical reasoning when uncertainty is communicated alongside AI outputs (Prabhudesai et al., 2023). While this evidence is not on R&D sensemaking, it indicates that awareness of AI-related risks can lead to better reasoning and may benefit the development of higher quality interpretations. Recent empirical work further shows that individuals react to AI suggestions in different ways. Di Prisco and Dello Russo (2026) distinguish between engaged acceptance, engaged refusal, unengaged acceptance and unengaged refusal, thereby separating the users' efforts of trying to understand AI outputs from accepting or rejecting them (di Prisco & Della Russo, 2026). For sensemaking in R&D this implies that AI may enhance interpretational activities through triggering critical reflection, but it may also weaken sensemaking if actors simply accept or dismiss AI output without challenging their assumptions.

AI in sensemaking: Negotiation

Another important aspect of the interpretation process is the negotiation of meaning among stakeholders to develop a sufficiently shared account (Maitlis & Christianson, 2014), an activity likely to be influenced by AI as well. Suri (2025) theorizes, that, instead of developing and negotiating

individual interpretations, stakeholders may shift to merely ratifying AI outputs (Suri, 2025). She argues that the availability of quick answers to complex questions can introduce a speed bias, which may pressure decision-makers toward premature actions before alternatives have been fully explored (Suri, 2025). An experiment on R&D investment decisions showed, that executives attributed a higher quality to AI-supported recommendations and favored them over human recommendations, indicating a risk of overreliance (Keding & Meissner, 2021). Similarly, Ben-Shahar et al. (2025) argue that AI can reduce information heterogeneity in board decision-making by presenting outputs as a single objective truth (Ben-Shahar et al., 2025). Suri (2025) further suggests a social confirmation bias, where repeatedly relying on AI as an institution pushes actors to be more likely to align with this new emerging source of consensus (Suri, 2025). This risk may be reinforced by the nature of generative AI, as those tools can produce seemingly plausible and coherent outputs that still contain factual inaccuracies (Kalota, 2024). In a sensemaking context, where plausibility is often favoured over accuracy (Weick, 1995), such AI outputs can become especially influential. Empirical evidence surfaces a related risk addressing creativity, when relying on AI outputs. Wadinambiarachchi et al. (2024) found that exposure to AI-generated inspiration in a visual design context reinforces design fixation and reduces variety of individual interpretations (Wadinambiarachchi et al., 2024).

AI in sensemaking: Building confidence on resulting actions

In addition to perceptions of uncertainty and plausibility, AI may also reshape how R&D teams justify decisions and build confidence for resulting actions. This requires viewing decisions and actions from a legitimacy perspective. Suchman (1995) defines actions as legitimate, when they are *"desirable, proper, or appropriate within some socially constructed system of norms, values beliefs and definitions"* (Suchman, 1995). This perspective is important, as legitimacy can have considerable organizational effects, and for some cases may even be associated with organizational survival chances (Ruef & Scott, 1998). In an AI context, legitimacy depends not only on outcomes, but also on decision process characteristics (Martin & Waldman, 2023). Factors influencing perceived legitimacy of algorithmic decisions include decision importance, degree of agreement with the outcome, use of morally questionable or arbitrary factors or possibilities to appeal to a human decision maker to review the decision (Martin & Waldman, 2023). Conceptual work further argues that transparency and explainability of AI outputs that help stakeholders to understand how a conclusion was reached, can alter legitimacy perceptions (de Fine Licht & de Fine Licht, 2020). Related work in healthcare and creative industries suggests that transparency, contextual understanding and human agency may shape the amount of trust users have in an AI collaboration (Daly et al., 2025). Therefore, the use of AI does affect decision justification not only by providing additional information, but it may also change the patterns of social acceptability and credibility of a decision.

In summary, these findings of directly relevant and adjacent literature suggest that AI may shape the way R&D teams handle uncertainty not only by determining which cues are relevant, but also by changing the perceptions of plausibility and legitimacy as well as processes of interpretational convergence. The following section 2.4 synthesizes these insights by connecting AI use to organizational sensemaking and derives research questions for the empirical part of this thesis.

2.4 Synthesis

In this chapter relevant findings from literature presented in sections 2.1, 2.2 and 2.3 are combined. First, a conceptual framework is derived from the proposed literature as the initial analytical basis for the empirical study. Then, related literature and concepts are used as a starting point to derive research questions for this study.

2.4.1 Tentative Framework

Based on literature and conceptualizations discussed in previous chapters, a conceptual framework meant to guide this thesis is derived. Figure 1 shows a graphical representation of this framework, consisting of collective sensemaking conceptualized as a process with outcomes of generative AI involvement mapped onto this process. The framework frames sensemaking as starting off with an incident triggered by uncertainty, leading the team to engage in collective sensemaking. This thesis distinguishes four interrelated sensemaking activities, including cue bracketing, interpretation, negotiation of meaning and action or decision justification. These activities may overlap and recur, rather than representing a rigid sequential process. Generative AI is conceptualized as a tool involved in this process, while four potential effects of its use on collective sensemaking process are highlighted. These effects concern the boundaries of search space, convergence or plurality of interpretations, debate and contestation as well as decision confidence and justification. While arrows indicate which activity each outcome is likely to affect primarily, they act as anchors rather than rigid causal relations, as in reality effects may overlap. This framework acts as the basis guiding the empirical study, but it is to be seen as tentative and is set to evolve with findings emerging from collected data.

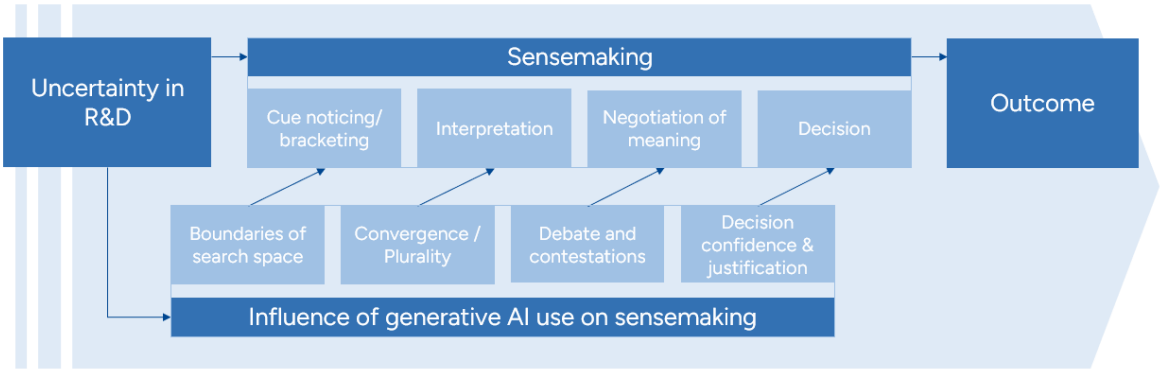


Figure 1: Tentative literature-informed framework for generative AI use in R&D sensemaking

2.4.2 Research Questions

Uncertainty is strongly linked to perception, implying that it matters how teams interpret what they do and do not know (Milliken, 1987). To make sense of uncertain situations, teams in organizations rely on bracketing cues, negotiate plausible interpretations and act, with a shared sense of meaning often being an important basis for coordinated action (Maitlis & Christianson, 2014; Weick, 1995; Weick et al., 2005). This process is likely to be increasingly mediated by AI. Algorithms can shape, which information is treated as relevant (Gillespie, 2013). From a sensemaking perspective, this suggests that AI may also have an influence on which cues get noticed. At the same time, research indicates two risks. At the individual level, decision-makers may put too much trust into and become biased toward AI recommendations (Keding & Meissner, 2021). On a collective level, AI outputs may be treated as an objective authority and contribute to a pseudo-consensus (Ben-Shahar et al., 2025). Further, whether AI-supported interpretations contribute to R&D decisions may depend on perceived legitimacy. Applying Suchman's (1995) concept of legitimacy to the decision context in sensemaking, this is the case when organizational actors perceive the use as appropriate within certain established norms and beliefs. Acceptance may be influenced by factors like decision importance (Martin & Waldman, 2023) or transparency and explainability of results (de Fine Licht & de Fine Licht, 2020). Much of the literature on AI in R&D emphasizes adoption, capabilities and outcomes, while the question of how generative AI is utilized in day-to-day meaning construction and justification under uncertainty remains scarcely explored. Literature surfaces a need for more empirical research on changes of behavioural patterns, judgement and justification practices around

human-AI interfaces in organizations (Keding & Meissner, 2021; Suri, 2025). This leads to the main research question of this Thesis:

How does the use of generative AI tools shape practices of individual and collective sensemaking in R&D?

An important component of collective sensemaking is the negotiation of shared, or sufficiently compatible understandings (Weick et al., 2005). In this process, agreement can be temporary and perceived plausibility on a narrative is often favoured over accuracy (Maitlis & Christianson, 2014; Weick et al., 2005). On the one hand, AI may broaden the available information base by processing large amounts of data and surfacing additional opportunities (Broekhuizen et al., 2023; Kakatkar et al., 2020; Truong & Papagiannidis, 2022), which may help to expand interpretations. On the other hand, AI may also supress interpretations and push premature convergence (Ben-Shahar et al., 2025). This tension was addressed in Sub-RQ 1:

How does the use of generative AI tools affect interpretive plurality in R&D sensemaking?

Debate and contestation can contribute to establishing collective meaning (Maitlis & Christianson, 2014) and shape interaction between leaders and stakeholders in the broader domain of sensemaking and sensegiving (Maitlis, 2005). Theory warns that AI-outputs may become a privileged reference, where they might undermine heterogeneity and information elaboration in group decision processes, thereby changing contestation practices (Ben-Shahar et al., 2025). One the one hand, management literature raises a need for critical thinking in order to challenge possible biases in AI outputs (Larson et al., 2024). On the other hand, research suggests that AI use might come at a cognitive cost of decreased critical thinking capabilities (Gerlich, 2025). Even though the contributions of Larson et al. (2024) and Gerlich (2025) are not directly related to R&D sensemaking, the studies suggest a tension between trust and contestation of AI outputs. The effect of those on collective sensemaking dynamics, and whether generative AI outputs are perceived a valid or even trusted contribution, must be examined more closely in an empirical setting. This leads to Sub-RQ 2:

What role does generative AI play in collective sensemaking in R&D teams?

In practice, when making sense of uncertainty, individuals often emphasize plausibility of stories over accuracy (Weick et al., 2005). AI may help managing informational uncertainty (Wu & Shang, 2020) while research in high-tech firms indicates that AI-adoption is positively associated with effective decisions (Ali et al., 2024). However, while AI supports data intensive decision processes, limitations lay in interpretability, causal reasoning and alignment with existing norms (Hao et al., 2025). Empirical research on R&D investment decisions suggests that decision makers often attribute greater trust and a higher perceived quality in AI recommendations compared to human advice (Keding & Meissner, 2021). At the same time, conceptual work raises the risk that AI may be relied on in high-stakes settings without formal delegation (Suri, 2025). In contrary, other research shows that individuals favour human forecasters over algorithms, when they witness an algorithm making unexplainable errors (Dietvorst et al., 2015). These partially contradicting findings suggest that generative AI use may alter the confidence levels of decision-makers. Sub-RQ 3 examined this relationship:

How does generative AI influence the practices under which R&D actors build confidence in decisions?

The four research questions were phrased to capture core activities around sensemaking in R&D and acted as the basis for methodological and analytical considerations.

3 Methodology

The following chapter features research design and setting, as well as a description of data collection, data analysis, ethical considerations, limitations and a brief discussion of alternative methodology considerations.

3.1 Research Design

This thesis was conducted as an exploratory qualitative study of two large industrial organizations in Sweden. The study examined how the use of generative AI shapes sensemaking under uncertainty in an R&D context. The dual case setting of this study features two established industrial firms with operations in Sweden. While both organizations maintain R&D activities in technology intensive contexts, they differ in their sector and market orientation. This setup allowed for exploration of practices in related, yet distinct settings. While the study does not claim statistical generalizability within the narrow setting of two cases, the two organizations provided access to varied industrial R&D settings. As the research questions focus more on interpretations, practices and legitimacy perceptions rather than measurable predefined variables, a qualitative design was chosen. This allowed for more faceted discoveries in the previously scarcely explored domain of how organizational actors describe the role of generative AI in R&D-related sensemaking. The study followed an abductive logic. An initial, tentative framework was informed by existing literature on uncertainty, sensemaking and AI in R&D and then refined by emerging patterns in the empirical data. The abductive approach is appropriate as existing theory does not yet fully explain how generative AI shapes sensemaking in R&D. The aim was therefore neither to test an existing framework deductively, nor to inductively build theory solely on the empirical data, but to develop explanations that consider both theory and empirical findings.

3.2 Research Setting

The study was conducted on two large industrial organizations with operations in Sweden. Both companies operate internationally within technology-intensive sectors and involve product development, manufacturing, automation solutions and service activities. Company A is a Sweden-based organization with global operations, focusing on the development, manufacturing and sales of products and solutions to both professional and consumer markets. Its operations include product development, manufacturing, automation and service-related activities. Company B is an international company with a parent company based in Japan and operations across Europe. It offers products, services and solutions in a logistics-related business-to-business context. Their operations involve development, manufacturing, sales, automation, services and digital solutions.

The empirical focus of this study was the R&D organizations of the two companies. Participants were recruited through access-based sampling by intentionally targeting relevant R&D roles within the access constraints. The final selection of participants represents a variety of R&D functions including senior management, innovation professionals, and engineering professionals across mechanical, electrical and software-related domains. Both companies had centrally driven AI initiatives and provided their employees with access to large language models in secure company environments. This made the cases relevant for examining how generative AI tools are used to handle sensemaking within uncertain, technically complex contexts in established, industrial firms. The cases were selected as they represent related yet distinct empirical settings. Both operate in technology-intensive, industrial contexts, while they differ in their addressed segments. Company A serves both professional and consumer-oriented markets, while Company B operates primarily in a business-to-business context.

3.3 Data collection

The empirical data was collected through qualitative semi-structured interviews with individuals from both case organizations. The purpose of the interviews was to understand how different actors across the R&D organizations encounter and make sense of ambiguous situations and whether generative AI tools play a role in these processes. Interviewing individuals with different roles allowed access to both technical and managerial perspectives on sensemaking in R&D. All interviews were conducted in online meetings in English and lasted approximately 50 to 60 minutes. With consent of the participants, the interviews were recorded and transcribed for analysis.

3.3.1 Recruitment and participant selection

Participants were recruited through access-based sampling, with the initial contact being established through the supervisor of this thesis, and additional participants being identified through snowballing. Within the access constraints participants were intentionally selected to represent a broad range of R&D functions. Senior managers were included as they are involved in strategic decision making around uncertain technical problems and the facilitation of corporate AI. Engineering professionals were included because they are directly involved in research and development work, where they regularly face technical uncertainty in practice across mechanical, electrical or software engineering contexts. Innovation professionals were included because of their involvement in R&D through close collaboration with both managers and engineers, for example in innovation workshops. One participant from Company B did not belong to the intended managerial or engineering contexts but was included because of involvement in R&D-related innovation workshops. The perspective was relevant because the participant had contributed to technical discussions with the help of AI despite not having a technical background. To maintain anonymity, this participant is subsequently referred to as an innovation professional. AI use was intentionally not a prerequisite for participation, as the research aimed to study AI use behaviour in practice. Including both frequent and less frequent AI users made it possible to identify boundaries of generative AI supported sensemaking, instead of reporting only positive cases.

3.3.2 Empirical material

The empirical data was collected through two rounds of interviews. The first-round interviews were conducted together with a fellow researcher as part of a related research project and consisted of 14 interviews addressing the use of generative AI in R&D more broadly, including the technology rollout processes, its justification and perceived legitimacy. Out of the 14 interviews, 2 interviews were excluded from this study, as the interviewee roles were not related to R&D and therefore not relevant for the sampling as described above, leaving 12 interviews included. The interviews were of relevance for this thesis, as questions were added addressing AI use in specific parts of the sensemaking process including information search, interpretation, decision support as well as trust, confidence and legitimacy perceptions toward AI-generated outputs. Since the scope of the first-round interviews was broader, only passages related to generative AI use in individual and team sensemaking in R&D, particularly reflections on perceived legitimacy, trust and verification of outputs were included in the analysis. The second interview round focused more specifically on sensemaking in R&D work. It consisted of 13 interviews, out of which 7 were follow-up interviews with participants from round one, while the other 6 were conducted with new participants. The interview guide of the second round was structured around three major parts. First, interviewees were asked to describe their role, recent projects and the types of problems and decisions they usually get involved in. Second, the interviewees were prompted to recall situations involving novelty, ambiguity, conflicting interpretations or decision-making under uncertainty. Third, participants were asked to reflect on whether generative AI played a role during these sensemaking incidents. If generative AI was used, follow-up questions

were aimed at understanding how it was used in activities corresponding to the tentative framework. When generative AI had not been used, participants were asked to reflect on factors constraining or limiting the use of AI. Even though the interview guide was designed to collect data around specific sensemaking incidents, the exploratory nature of the research intentionally allowed for deviation from this approach. While some interviews focused on specific incidents, others evolved around broader reflections of participants on their AI use behaviour around uncertainty and sensemaking in R&D. This approach allowed to follow new themes that emerged during the interviews, while the central themes were maintained across all interviews. The interview guide is added in Appendix A.

In total, 27 interviews were conducted within the broader research process. Out of these, 25 interviews with 18 individuals were included in this thesis, while two first round interviews were excluded as the participants' roles were not relevant for the focus around R&D. Company A was represented in 9 interviews with 7 individuals, and Company B was represented in 16 interviews with 11 individuals. An overview of the interviews included in the empirical material is attached as Appendix B.

3.4 Data Analysis

This study used an abductive approach to the qualitative analysis. Existing literature on uncertainty, sensemaking and AI in R&D informed the initial conceptual framework. However, the framework was not treated as fixed but evolved with the empirical findings in combination with literature. This is consistent with the concept of *Systematic combining*, where a theory-informed framework directs the collection of empirical data, which is then utilized to evolve and potentially redirect the framework (Dubois & Gadde, 2002). An abductive approach was appropriate in this situation, where existing theory provided relevant understanding for sensemaking and AI use in R&D, but did not yet fully explain how generative AI shapes sensemaking in practice. According to Timmermans and Tavory (2012), abduction is particularly suitable to develop theoretical explanations for empirical findings that existing concepts do not fully explain (Timmermans & Tavory, 2012). The goal of the data analysis was therefore not deductive proof of the tentative framework or an inductive construction of theory, but rather to build a plausible and conceptually grounded explanation of how generative AI shapes sensemaking in R&D practice, considering both existing theory and discoveries in the empirical data.

Interview recordings were transcribed using the built-in transcription function of Microsoft Word and then manually reviewed and corrected. During this process, the transcripts were edited to improve readability, which included the removal of filler words, repeated words and parts of the conversation unrelated to the interview, such as technical issues during online meetings. Other than that, the participants' statements were preserved in their original wording and meaning. In total, the coded material consisted of approximately 22 hours of interview material or 9500 transcribed sentences.

In qualitative analysis, coding refers to systematically assigning labels to text passages in order to identify broader recurring patterns related to the research questions. Coding was done in MAXQDA in two rounds. The initial round of coding was done using a codebook that reflected the tentative framework, including codes related to uncertainty, the sensemaking stages, interpretation, team discussion and decision confidence. The coding process remained open at this stage and emerging codes were added to this version of the codebook. After the first round of coding, the initial and emerging codes underwent a review and reorganization. This included the combination of redundant codes, separation of broad codes and reformulation of unclear codes. The new codes were structured around four themes that represented sensemaking activities participants had frequently described, including exploration and information search, interpretation and articulation, preparation for team discussion and confidence and verification. Other themes captured AI use cases, further conditions relevant to generative AI use or codes that could not be clearly assigned to one of the four sensemaking

activities. The second round of coding utilized the revised codebook for all interviews. A formal comparison across cases or roles was not made. Even though company and role context were considered during coding, the analysis suggested that differences in AI-supported sensemaking were more related to the specific task domain, task stakes and verification possibilities than to a specific company or role.

The coding was conducted by the author. As a supplement, ChatGPT was used to identify potentially left out passages relevant to the codebook, that may have been missed during manual coding. AI-generated suggestions were reviewed by the author and added to the coding, when analytically appropriate. Decisions and responsibility remained with the author, as ChatGPT was only used as a support tool and not as an independent coding authority. Transcripts were anonymized and passages including specific identifiable incidents rather than broad reflections were removed for this step. To provide transparency, the coding process was described in detail and both the initial and revised codebook were included in Appendices C1 and C2. Furthermore, three coding examples, presented as excerpts from coded interview segments, were included in Appendices D1, D2 and D3.

3.5 Ethical Considerations

All participants were informed about the purpose of this study and participated voluntarily. Interview recordings were made with the participants consent. Company and participant names were anonymized, and industry and roles descriptions remain at a level to avoid identification of participants. The analysis and the presentation of the results within the thesis avoids disclosure of confidential information.

3.6 Limitations

The study has several limitations. First, as common for single or dual case studies, one limitation is the limited generalizability across industries, organizations and R&D domains. The study therefore does not claim statistical generalizability and should instead be seen as an exploratory and analytical contribution. In the scarcely researched area of generative AI in R&D sensemaking, this approach was particularly valuable to discover practices and patterns, that should be addressed in future research on a larger scale.

Second, participants were recruited based on access and through snowballing, which may introduce bias toward participants that were available and willing to contribute to this research. This may have led to individuals less engaged or more sceptical toward AI use to remain underrepresented in this study. To mitigate this, participants were intentionally selected across management, innovation and engineering roles. Further, AI use was not mentioned as a requirement for participation.

Third, the two interview rounds were not based on the same interview guide and therefore do not relate to the research questions equally. The first-round interviews, conducted as part of a related yet separate research project, had a broader focus around generative AI use practices and legitimacy perceptions. Round two was focused more specifically on generative AI use in R&D sensemaking, limiting comparability across the two rounds. However, the first-round interviews were still included in the analysis, as they contained passages relevant to the present study, adding empirical breadth.

Fourth, due to access and time constraints, the data collection relied on retrospective reconstruction and reflection of AI-supported sensemaking instead of observation of sensemaking in real-time. This may have introduced a bias toward emphasizing successful cases, leaving problematic practices underreported. Further, participants may have forgotten details or simplified sequences of events. Similarly, incidents discussed during the interviews were mostly reconstructed individually from a singular perspective rather than by multiple stakeholders. Data on collective sensemaking

therefore remained limited to participants' descriptions and did not fully capture discussion dynamics. However, the perspectives provided by participants were still analytically valuable and provided understanding of legitimacy perceptions and individual practices within sensemaking on an individual and team level.

Finally, due to lack of access to a co-researcher, the coding was conducted solely by the thesis author, introducing a risk of interpretative bias. This was mitigated by two rounds of coding with iteration of the codebook. Further, the coding process as well as the final version of the iterated codebook were documented transparently, and findings were presented including examples in the form of interview quotations.

3.7 Discussion of alternative approaches

Other options of methodology were considered for this thesis work. First, a quantitative study would have allowed to capture more breadth and potentially gain more generalizable insights. This approach would have further allowed to quantitatively measure the object of investigation, the influence of generative AI on sensemaking processes, and to compare the influence of different factors and the severity of effects. However, such an approach would have required grounded hypotheses and a clear map of distinct dependent and independent variables, which the available literature base did not offer. Additionally, a quantitative setup may not have been sufficient to cover all subtle facets of behavioural patterns along the sensemaking process. This is where the qualitative approach holds its strength, as it allows to not only observe these patterns but to also analyse underlying mechanisms and team dynamics. As little was known on AI in R&D sensemaking, a qualitative approach allowed for a broad exploration in this field.

One qualitative approach may have observed sensemaking happening in real time. This could have been done by joining team meetings, where stakeholders make sense of an incident by searching information, debating interpretations and concluding actions. Combined with individual interviews of stakeholders along the way, this approach would have allowed for rich data collection. In addition, as sensemaking unfolds in real time, debates may have been observed as they happen, giving less room for bias compared to a reconstruction of events solely based off interviews. The approach would likely have provided richer data on discussion dynamics. However, it would have required observation of multiple teams over an extended period, which was likely to exceed the timeframe allocated for data collection within this thesis. Furthermore, the logistical effort of coordinating and joining such meetings within the given time frame was out of scope for this thesis.

Finally, another viable option considered was conducting interviews with multiple individuals who were mutually involved in shared sensemaking incidents. This approach could have helped to draw a more faceted picture of certain sensemaking scenarios by incorporating multiple perspectives of the same incidents and thereby offering rich insights into debate practices and collective sensemaking patterns. However, after this approach had initially been pursued, difficulties emerged for three reasons. First, pinpointing certain incidents without explicitly disclosing on company internals posed a reasonable challenge for the interviewees. Second, limited access to multiple stakeholders per incident posed a risk given the short timeline of this degree project. Third, in the few cases with multiple stakeholders available, the ability to recall details among interviewees varied greatly, risking bias in the data. For these reasons, the initial methodology shifted to a more individual approach, maintaining an emphasis on critical incidents while allowing more room for individual reflections.

The chosen approach for data collection was based on individual interviews, aiming to trigger reflections on sensemaking episodes on both individual and team level. This balanced the quality of the collected insights with an efficient research process and allowed to acquire a solid data basis within the constraints of a master thesis' limited scope.

4 Results

This chapter presents the results of the data analysis, structured according to empirically identified themes that emerged from the revised codebook during the abductive coding process, reflecting areas of sensemaking activities in which generative AI played a role. The four areas were exploration and information search, interpretation and articulation, input to team discussion as well as confidence and verification. While the tentative framework guided the first round of coding, the initial codebook was iterated according to the empirical findings. The revised codebook was then reorganized to reflect main areas of activities participants described in AI-supported sensemaking. These activities correspond to elements of the tentative framework, but they should not be seen as a fixed linear sequence. This structure was considered appropriate because the empirical evidence indicated different levels of AI reliance and use frequency across different sensemaking activities, making these activities a central organizing condition for involvement of generative AI. Throughout the interview data, generative AI was most frequently described as supporting exploration, information search and interpretation, while the impact on team discussions and activities that involve building confidence around decisions was more indirect and conditional on other factors. Presenting results around the emerging themes therefore shows how generative AI entered different activities of sensemaking in different ways. An alternative structure based on company, cases or roles was considered less suitable. Even though these factors were considered during the analysis, patterns emerged more strongly around specific activities, task domains, task stakes and possibilities of verifying the AI output. In Table 1 findings were summarized and organized by the main emerging themes. The subsequent chapters were then used to describe findings in more detail, including examples and references from the coded segments.

Table 1: Summary of findings from the coded interview data. 'AI' refers to generative AI tools

Sensemaking activity	Patterns
Exploration and Information Search	• AI accelerates information search • AI broadens information search space • AI supports new domain entry • AI offers a starting point • Boundary condition: Knowledge accessibility and specificity
Interpretation and Articulation	• AI generates additional perspectives • AI serves as a sparring partner for developing interpretations • AI supports articulation of content • Boundary condition: Access to relevant contextual knowledge
Input to Team Discussions	• Consensus through team discussions remains essential • AI enters discussions more indirectly through individual preparation • AI may lower the participation threshold for some individuals
Confidence and Verification	• Task and decision ownership remains human • Confidence in AI-supported decisions is ambivalent and likely depends on task stakes and verifiability of outputs

4.1 Generative AI in exploration and information search

One of the clearest patterns of generative AI use in R&D sensemaking emerged around exploration and information search activities. Generative AI was repeatedly described as a strong support in exploratory sensemaking activities for quick orientation and background research in new domains or as providing a starting point for a given problem. The data suggests that generative AI supports exploration and information search in multiple ways.

First, generative AI was repeatedly described as accelerating information search. This pattern appeared in interviews with 13 out of 18 participants, either through explicit statements or through examples where generative AI allowed them to find relevant information more quickly. Some reported that they use large language models like an advanced search engine, replacing traditional web search. One participant described the development like this: *"Before AI, I would go online, Google things, read a lot of articles, and so on, which takes a lot of time. […] Now that part is much faster"*. On top of that, generative AI was often described as useful when searching for relevant segments in known information, such as codebases or user manuals, significantly accelerating an otherwise manual process. An R&D manager among the participants reflected that *"[AI] speeds up the efficiency of engineers, because before they had to look for information across different folders, different Teams channels, and so on. Now they can get that information through a prompt, through a chatbot"*. Participants achieved this increase in speed by prompting an LLM to either create summaries of information, or to point to specific relevant sources or segments within search spaces accessible to the tool. The data shows that generative AI allowed users to scan through available data more quickly.

Second, participants often reported to use generative AI to broaden their information search space. This was the case, when it enabled a participant to use information around a given area of interest that was previously inaccessible to them or that they simply did not know about before. For example, one participant described a case of performing comprehensive research on computer chips for a specific use case. In this case, an LLM helped the participant to expand the search space beyond datasheets available in familiar languages and identify relevant Chinese suppliers. Another participant described regularly scanning academic publications for new technology relevant to their own domain. These papers, often published by leading experts in highly specialized fields, would previously have been out of scope because the participant lacked expertise in those domains, leaving potentially relevant information undiscovered. A participant in engineering summed up the role of generative AI as broadening the search space like this: *"I still bounce my ideas with my team members as well, but the AI plays more of a role in helping me find out what is available"*.

Third, the data showed that participants were supported by generative AI not only to search within new or unfamiliar domains, but also to enter them more effectively. One participant reported that an LLM supported them selecting among less familiar programming languages for a large project, without requiring a deep dive into each language. The participant further used the LLM to translate concepts and ideas from familiar programming languages into the new domain. Another example was provided by a participant integrating LiDAR technology into a prototype, where an LLM supported them in finding libraries and segments of the codebase relevant for this task. A third participant described an example of choosing among unfamiliar image, data or file formats related to computer vision, stating that *"AI lets you expand your domain much more quickly"*.

A related yet distinct pattern was, that generative AI was often described to provide a starting point into a given task or problem. Participants talked about turning to generative AI to ease their way into a project in early stages. While the previous pattern was about entering unfamiliar knowledge domains, this pattern emerged around generative AI producing an initial direction, that participants could further refine. An engineer described it like this: *"It's mostly like these early questions when I don't really know which direction I'm going, need a better understanding or explanation of some concepts, or need a starting point, then I go to AI"*. Another participant provided

an example, where they used generative AI to compare solutions of competitors to a given problem in their market, to *"understand where [they] should dig in deeper later"*. Similarly, generative AI was described as useful for providing skeletons for presentation structures or for software functions that were to be integrated into a larger codebase.

The usefulness of generative AI in exploration and information search was still described to be constrained by certain conditions. One boundary condition was the accessibility of relevant knowledge. AI was described to be particularly useful for information that was publicly available or in a search space accessible to the tool. However, a lack of access to company specific data was described as a major limitation. One participant described the issue like this: *"Copilot is in a kind of closed environment and not really able to search inside our own data. It is just searching outside. And that is what I am missing. I am totally missing a search engine that can search internal data"*. Another participant described the usefulness of generative AI for exploratory activities as task dependent: *"If it is a more general task, then an LLM is quite good to start with. But if it is a more specific internal task, then I think a colleague is always better"*. This suggested that the role of generative AI in exploration and information search was stronger for more generic tasks with publicly available information, and weaker for cases where cues were part of organizational knowledge inaccessible to the tool.

4.2 Generative AI in Interpretation and Articulation

Beyond initial information search generative AI was also described to support interpretation and articulation of given problems and tasks. While the results presented in the previous section showed how participants used generative AI to find information and orient themselves in new domains, the data suggests that it was also used to interpret the information at hand. At this stage generative AI was not primarily used to generate final answers, but rather to support participants in achieving a clearer understanding of a problem in three main ways: By generating additional perspectives, by helping participants articulate given issues more clearly, and by being utilized as an interpretative sparring partner.

First, generative AI was repeatedly described as a useful tool generating new perspectives around a given problem. Several participants reported to regularly be using generative AI to generate alternative solutions or viewpoints that they had not considered themselves. One participant stated, that *"you can always ask and see what new ways there are of understanding the problem, or what new possible solutions exist that you did not think of yourself"*. Sometimes the generated perspectives even altered established organizational habits. One participant provided an example, where an LLM generated an alternative interpretation of a norm, stating that *"there was actually another valid way of seeing the issue, but we were so used to walking one path that we had not really noticed it"*. Another example was provided by an R&D manager, who reported to specifically use generative AI to roleplay certain perspectives and generate new viewpoints: *"For example, I might say, 'now you are a critical CTO. Please review this material form the eyes of a critical CTO'"*.

Second, generative AI was used as a sparring partner supporting the development of interpretations before involving other people. Several participants did not only use generative AI as a tool to get single answers but rather described an iterative process of going back and forth on a certain topic. For example, one Engineer described an AI tool *"as a sounding board to bounce ideas back and forth and get arguments and numbers"*. Others compared generative AI use to getting a second opinion from an additional colleague. One participant described it like this: *"Just like I might have a colleague who is an electrical engineer and who has a different input or a different take on a project"*. The role of generative AI as a sounding board or sparring partner was described to be particularly valuable because it was constantly accessible. One participant explained that they value pair programming with *"somebody to think aloud with"* and AI gave them *"constant access to that"*. *G*enerative AI was not only used for confirmation, as some participants stated explicitly that they used

it to challenge their own ideas. In this role, generative AI supported problem interpretation by helping participants to test and refine their assumptions before turning to a colleague or team.

Third, the data showed that generative AI was not only used as a sparring partner to develop and refine own interpretations, but also as a support in articulating them in a suitable way for others. For example, participants reported generative AI support in translating technical understanding for a non-technical audience. An R&D manager shared an observation of an engineer within their department: *"One of my consultants here is a very good engineer, and I realized the other day that he has started to use AI more and more to make things more understandable for others. He is a very skilled engineer, but he is not so keen on speaking in that way"*. In a broader sense, generative AI was described to be useful for framing content for certain receivers. One participant described it this way: *"I put in the content, but I would like to present it in a shorter format and make it more suitable for this kind of receiver"*. Others described using generative AI to add rhetorical or argumentative strength, for example by transferring an argument to a *"management level rhetoric"* or to *"make it fluffier"*.

An important boundary condition mentioned repeatedly was the dependency on organizational context for full understanding of a given problem. The capability of generative AI to support problem interpretation through generating perspectives, iterating ideas and communication of certain issues was therefore constrained by its access about specific organizational or tacit knowledge. One participant described the issue like this: *"I still cannot fit in every piece of knowledge that I have. I have been working in this department for seven years. If I could give it every interpretation and every bit of knowledge from those seven years, then it could probably deliver enormous value"*. Even though generative AI was often used to generate additional perspectives, human judgement remained essential to assess whether those perspectives were appropriate in the organization's broader context. Several participants also emphasized that experienced colleagues remained superior advisers when interpreting problems in the organizational context. An R&D manager among the participants put it like this: *"If it is a more specific internal task, then I think a colleague is always better"*.

4.3 Generative AI as input to team discussion

The previous subchapters showed that generative AI reportedly supported information search and interpretation, its role in sensemaking in team discussions was more limited. Participants rarely described collective generative AI use during team meetings and emphasized that AI does not replace the need for negotiating with colleagues and building consensus through team discussions. Instead, generative AI entered team discussions more indirectly through individual AI-supported preparation. Generative AI was reported to rather be used prior to instead of during meetings to help participants prepare informed contributions to given topics.

In contrast to the rich findings around information search and individual interpretative work, generative AI was rarely described to be collectively prompted during team discussions. Instead, it was used in a more indirect way by supporting individuals in preparatory work before meetings. One participant summarized this pattern after being asked what role AI plays in team discussions: *"For sure it plays a role there. But mostly in preparation and not so much in the actual meeting later on"*. Two forms of generative AI support in preparatory work emerged from the data.

The first form concerned the preparation of decision material. For example, one R&D manager described a recent strategic decision about the sensor setup of a product, where generative AI was utilized to collect arguments for each sensor type. The participant explained, *"AI was used to define the different alternatives and collect pros and cons of each alternative. So, it was used to prepare presentation material that could then be used in the decision meeting"*. However, the AI did not make the final decision, as the participant pointed out: *"That was more a matter of discussion […] and deciding on a way forward"*. Building on the practices of individual information search and

interpretation as described in sections 4.1 and 4.2, a second form of AI-supported decision preparation concerned the refinement of individual contributions. Participants used generative AI to test and refine ideas before bringing them to the team. One participant described using it in that way, to avoid a weak contribution to the team: *"I do not want to go into a meeting proposing something and then have everybody immediately shoot it down because I did not research enough"*.

While generative AI was reported to be used in preparation for team discussions, convergence was still described as evolving around human discussion. Participants repeatedly emphasized that discussions with the team remained essential to align on a shared understanding and to arrive to a decision on the right way forward. One participant explained that confidence around a decision was built, when *"the team, or all members of the team, draw the same conclusions or roughly the same conclusions"*. Another participant described that consensus evolved by *"getting the right people into the room, […] and then try putting the facts on the table and have a real discussion"*. In these cases, generative AI could inform a participant's understanding, but the AI output itself did not carry much weight in the discussion. One engineer described it like this: *"When you go into such a discussion, you are expected to know the facts. Saying that an LLM said this or that does not help. […] If you are in an argument and someone says they got it from AI, then it is almost worth nothing"*. Participants repeatedly reported that AI-supported discussion input only became credible when it was understood and owned by the person bringing it into the discussion. In these cases, credibility was dependent on the contributor's expertise rather than the AI output. One participant explained that they valued a colleague's contribution because *"he has shown over and over again that he knows what he is talking about and that he delivers. If I want to know the answer from Claude or ChatGPT, I can just ask them myself"*.

In one case, AI-supported problem framing appeared to affect who could contribute meaningfully to team discussions. The example came from the context of an innovation workshop, where an AI agent was embedded into a problem-solving tool. Reformulating a problem with the help of the AI tool lowered the threshold of participation in the technical discussion for a participant, despite not having a technical background: *"It was easier for me to feel that I could actually contribute. […] And there I absolutely found ideas that challenged their thinking"*. However, the role of the AI component cannot be clearly separated from the context of the used problem-solving tool and the described example was a workshop situation rather than an actual operational project meeting. The case should therefore be seen as an example suggesting how generative AI may lower the threshold for participation in team discussions for certain individuals.

4.4 Confidence and Verification in AI-supported sensemaking

As described in previous sections, participants described generative AI as conditionally useful in supporting sensemaking during certain activities, such as information search and problem interpretation. However, the data showed that AI support did not automatically lead to higher confidence in sensemaking outputs. Generative AI on its own was not described as trustworthy, instead it required verification and interpretation through human expertise to build confidence in its output. In this subchapter findings are summarized that concern how confidence is built around AI-supported sensemaking outcomes.

First, participants repeatedly reported that generative AI-outputs needed verification to be used in discussions or decisions. The responsibility toward tasks and decisions remained with the human and fact checking was therefore described as essential to maintain one's own credibility. One participant emphasized, that there *"must be an accountable human in the process"*, while another one stated regarding their own output: *"I need to own it. I need to be responsible for it"*. This was especially relevant for high-stakes tasks or work related to certification. One participant summarized it like this: *"if you really need something to be correct, you always need to check it"*. Verification was described to be done in different ways, depending on the type of output. For example, one participant

described a situation where they used an LLM to find information in a component's user manual, to then check the specific paragraph where the required piece of information occurred. A different example was provided by an engineer, who stated to check AI-generated code in the same way as they would review code of junior developers. Another engineer emphasized that AI-generated output needed to be interpretable by a human reviewer: *"We use the LLMs to produce source code we can read, because it's not ready yet to produce something we can just trust"*.

The confidence in generative AI when supporting decisions was different throughout the participants, and positive as well as negative effects were reported. For several participants generative AI increased their confidence by improving the basis for judgement in decisions. For example, it could provide feedback, support information search or generate additional perspectives, as described in chapters 4.1 and 4.2. When asked whether generative AI had an influence on their confidence in decision-making, one participant stated: *"Absolutely. And that is purely because I have more information and can get through information faster. […] I get more confident in what I do, and I get better quality in my work"*. Similarly, another participant described that they chat with an LLM tool *"to get a bit more confidence in something",* because they sometimes *"realized that it shows [them] something they have not thought of"*.

At the same time, some participants reported that the use of generative AI outputs decreased their confidence in decisions as it surfaced new uncertainty. These uncertainties were mainly caused by hallucinations, prompt sensitivity and agreeability of LLMs. Several participants emphasized that outputs depend on how the question is asked and highlighted a risk of the tool being overly agreeable with the user. One Senior Manager pointed out that LLMs are *"typically not that good at challenging"* and *"they often want to please you a little bit"*, while a Software Engineer stated to be *"scared of the AI just getting into a positive feedback loop for no reason"*. It was repeatedly reported that LLMs could easily be steered into certain directions, which could happen consciously but also out of the users notice. One participant summed this issue up like this: *"Everyone knows that you can get almost anything out of it if you try"*. A related source of uncertainty concerned hallucinations. Participants repeatedly reported that AI often generated outputs that appear plausible but are incomplete, oversimplified or factually incorrect. One participant gave an example of an AI-generated presentation for a workshop, where complex issues were overly simplified into strong claims: *"I think that was quite clear that it was making strong points out of maybe not that much"*. Another participant reflected on a situation, where they involved generative AI in troubleshooting for a technical component. In that case, the output made a credible impression because it included references but turned out to guide them in the wrong direction: *"I should have fact checked. That is obvious. But since it provided a lot of references, it felt like the proof was already there"*.

Whether confidence in generative AI-supported decisions was increased or decreased varied mainly with task stakes and the possibility to verify outputs. Participants reported to be more cautious when correctness of outputs mattered more. One participant stated, that *"you need to be aware of what kind of questions you are asking and how important the correctness of the answer is to your project"*. Another participant highlighted the difference on generative AI use behaviour dependent on task stakes across domains and development stages: *"I think you should use it in different ways. I think in validation, you really need to trust the data. Otherwise, you draw conclusions about the wrong thing. When it comes to idea generation, it's a creative phase. And you would like it to be crazy"*. Further, confidence was built easier when results could be verified, which was described to also depend on the user's domain expertise. Participants repeatedly emphasized that users with a higher level of domain knowledge are more capable of judging the output. One engineer highlighted it like this: *"The one who interprets the AI's result must have experience. If the person doesn't have experience, then you will believe what you see"*. Another participant directly explained how they perceived expertise changing risk and confidence levels regarding generative AI hallucination: *"In a domain where they are experts, they see hallucinations as a key feature. In a different domain, […] if you lean too much on hallucinated output and cannot filter it, it becomes a risk"*.

5 Analysis and Discussion

This chapter discusses the empirical findings in relation the research question and literature. Section 5.1 addresses the main research question, taking a broader view on the conditions under which generative AI enters sensemaking. Sections 5.2, 5.3 and 5.4 address the three sub research questions concerning interpretative plurality, collective sensemaking and confidence in AI supported sensemaking. Section 5.5 discusses a broader implication of generative AI on uncertainty in R&D.

5.1 Generative AI is used in R&D sensemaking conditionally

The tentative framework positioned generative AI as a technology that could have influence across individual and collective sensemaking activities in R&D. Before examining implications of generative AI for specific activities, this section starts by taking a broader view on AI's role in sensemaking, thereby addressing the main research question:

How does the use of generative AI tools shape practices of individual and collective sensemaking in R&D?

Throughout the cases studied, generative AI was described as useful supporting activities such as information search, problem interpretation or individual preparation for discussions. At the same time, it was less often described as directly entering collective sensemaking or team discussions. This suggests that generative AI did not generally replace human sensemaking in R&D but instead entered sensemaking practices conditionally. This subchapter provides an overview of these conditions, which are elaborated in more detail in subsequent sections. A graphical representation of these conditions is visualized in Figure 2.

First, generative AI involvement differed across sensemaking activities. In the interviews, generative AI was most frequently associated in sensemaking activities, where participants entered unfamiliar domains, searched for information, framed and interpreted problems and prepared material for further team discussions. The role of generative AI appeared weaker in activities around collective negotiation or building confidence around consequential decisions.

A second, partially related condition was the social mode of sensemaking, determining whether actors engaged in sensemaking individually or collectively. Participants described using generative AI more often individually, for example for searching information, exploring alternative perspectives or testing and refining ideas. In team settings, AI-supported inputs were described to be less credible unless owned by a human contributor. Within the studied cases, generative AI therefore appeared to influence collective sensemaking mainly indirectly, by shaping what individuals brought into the discussion rather than actively contributing to it as an independent team member. This does not mean that generative AI did not affect collective sensemaking, but its influence appeared to focus mainly on individual preparations before team discussions.

The third condition concerned task characteristic, including both task stakes and task domain. Participants described generative AI use as more acceptable in low stakes tasks such as exploration, drafting or idea generation. In relation to higher-stakes tasks, participants stated to be more cautious, especially in situations when correctness mattered. At the same time, AI and particularly LLMs, were described to be more useful for tasks that were text-based, such as coding, searching for information or summarizing text-based material. It appeared less useful in domains that could not be fully expressed in text, such as mechanical engineering or CAD-related work. These findings suggest that perceived appropriateness of generative AI was partially conditional on the potential consequences of relying on insufficient outputs, but also on whether the task domain could be appropriately supported by an AI tool.

The fourth condition was about the contextual embeddedness of the tool. Participants repeatedly described generative AI as more limited, when sensemaking involved tacit or organizational context rather than generic knowledge. This suggests that generative AI's perceived usefulness in R&D sensemaking was partly conditional on how much relevant contextual information was available to the tool and whether users were able to provide this knowledge.

Finally, the fifth condition concerned verifiability and human expertise. Throughout the interviews, participants repeatedly emphasized the importance to evaluate generative AI outputs through human judgement. A human actor taking ownership of AI output appeared essential for its credibility. When outputs could be verified, generative AI appeared useful for broadening the information search space and interpretive plurality. When outputs were difficult to verify or when the user's expertise was insufficient to reliably spot errors, generative AI could instead introduce new uncertainty regarding its output's reliability.

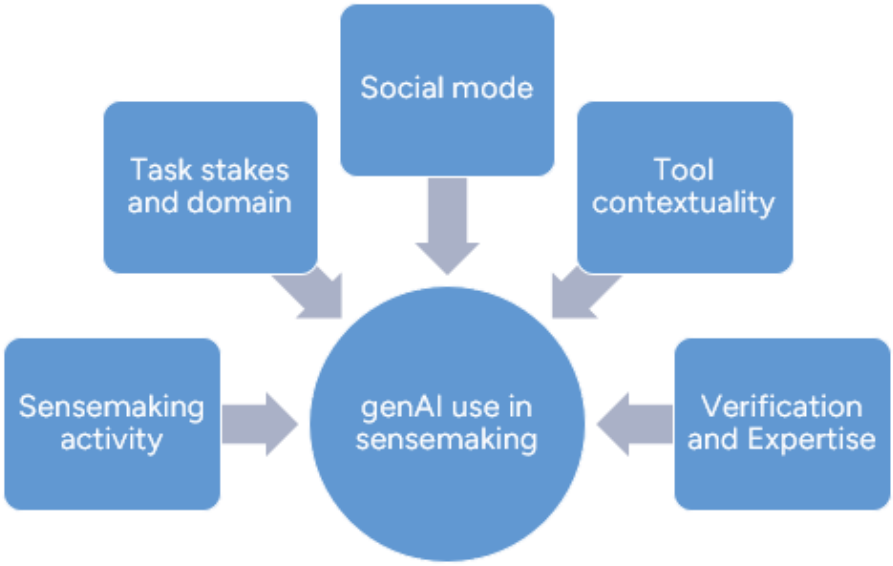


Figure 2: Conditions influencing generative AI use in R&D sensemaking

Taken together, the findings do not show that generative AI generally improves or weakens sensemaking in R&D. They rather indicate a more nuanced dependency of generative AI use in sensemaking on various conditions. The conditions are not treated separately in the following subchapters, as in practice they often overlapped. Instead, they act as boundary conditions for the discussion of how generative AI affected interpretative plurality, uncertainty, collective sensemaking and decision confidence throughout the study.

5.2 Generative AI may add perspectives if prompted to do so

According to Weick, actors engaging in sensemaking build plausible interpretations from collected cues (Weick et al., 2005). In uncertain R&D environments, some situations may allow multiple plausible interpretations of the same information, which actors then negotiate in an effort to converge around sufficiently compatible interpretations (Maitlis & Christianson, 2014; Weick, 1995). This raised the question, whether generative AI affects the plurality of competing interpretations and how this affected sensemaking itself. This problem was captured in sub research question 1 and addressed in this section.

How does the use of generative AI tools affect interpretive plurality in R&D sensemaking?

The literature suggests competing directions regarding this question, although the discussed studies originate in different empirical contexts. On the one hand, research on AI in innovation management emphasizes that AI can scan large amounts of data and may help overcome human information-processing constraints (Haefner et al., 2021; Truong & Papagiannidis, 2022). Given the increase of available information, this suggests that AI may as well support interpretative plurality. On the other hand, Gillespie (2013) shows, that algorithms may influence which information is treated as relevant (Gillespie, 2013). Even though this research is framed at algorithms more broadly rather than on generative AI alone, this contribution is relevant as interpretations may depend on what information is treated as relevant by the user. A related concern is suggested by Wadinambiarachchi et al. (2024), who find that AI-generated suggestions in a visual design context favour the users fixation around the AI suggestion (Wadinambiarachchi et al., 2024). Even though this should not be transferred directly into an R&D sensemaking context, the research highlights a relevant risk of users converging on an AI-generated solution without fully exploring other alternatives.

The findings of the study reflect this tension. Participants described to have used AI to generate additional viewpoints or act as a sparring partner to challenge ideas and assumptions. However, at the same time other participants described that generative AI helped them polish or fluff out ideas. This did not necessarily broaden the interpretation, but it rather appeared to add theoretical and argumentative strength to an interpretation when communicating it to others. The data therefore suggested that generative AI could support interpretative plurality, but this depended on whether the user prompted it to do so. The data is consistent with the argument made by Haefner et al. (2021) and Truong and Papagiannidis (2022) that AI can increase information processing capacity. However, the assumption taken during this thesis that informational breadth automatically translates to interpretative plurality was not supported by the data and these two concepts should therefore remain analytically distinct. The findings suggest that the extent to which more information access leads to richer interpretations appears to depend on how users interact with the tool. This also extends the argument of Gillespie (2013) that algorithms shape relevance for the case of generative AI. The findings support this view for some cases, where generative AI reportedly reinforced the initial idea when asked to elaborate on it or even to improve it. However, the relevance was not shaped by the tool alone, but also by the way it was prompted by the user. Depending on the interaction of user and tool, the same tool could be used to broaden perspectives on an issue, to narrow it down or to polish the initial direction. The findings also provide a nuanced view on the concern about premature convergence. The empirical data suggests that generative AI may elaborate on earlier user inputs in ways that reinforce an initial path, which users described as the tool being too agreeable. However, the data also showed that this behaviour of the tool may be actively avoided through adequate prompting technique.

Overall, the findings suggest that generative AI appeared not to increase or decrease interpretative plurality in R&D sensemaking by itself. In the studied cases, the effect seemed to be conditional on the interaction of the user with the tool. Generative AI could support divergence when used to generate alternative interpretations or challenge assumptions, but it could also support convergence when prompted to reinforce an existing interpretation. This indicates that access to more information through generative AI did not automatically increase the richness of interpretations. This condition appeared to depend on whether the users were seeking divergence or confirmation.

5.3 Generative AI support for collective sensemaking appears indirect

Collective sensemaking includes actors negotiating interpretations to develop a shared understanding that supports action. While the previous section focused on whether generative AI expands or narrows interpretations, this section examines the question, to what extent it enters the social negotiation processes of collective sensemaking. This issue was addressed in sub research question 2:

What role does generative AI play in collective sensemaking in R&D teams?

Literature highlights the social factor in collective sensemaking. Maitlis (2005) emphasizes that sensemaking is shaped by interactions of mutual sensemaking and sensegiving among stakeholders and leaders (Maitlis, 2005). Gioia and Chittipeddi (1991) describe the practice of sensegiving in strategic change as a cyclical process where organizational actors first make sense of change for themselves and then engage in sensegiving to influence others (Gioia & Chittipeddi, 1991). Even though their study is focused on strategic change initiation, rather than R&D team discussions, the concept shows how individuals prepare interpretations they later communicate to others. The empirical findings suggest that generative AI may have entered this process partially. The social process of negotiating meaning within a team did not appear to be directly replaced by generative AI. However, AI tools helped individuals prepare their input for discussions by structuring arguments, iterating and refining their ideas or preparing decision material. In these cases, generative AI could be seen as a sensegiving support tool used prior to the discussion that helped individuals to enhance the argumentative and rhetorical strength of their communication. The social practice of sensegiving, however, appeared to remain owned by humans.

Ben-Shahar et al. (2025) warn that, in corporate board decision making, AI outputs may become treated as an objective authority in discussions. They argue, that this may reduce heterogeneity of information, suppress the elaboration of information and ultimately lead to what they call pseudo consensus (Ben-Shahar et al., 2025). Although their framework addresses board-level decision making rather than R&D sensemaking, it provides a relevant point of examining whether AI becomes an authority in collective interpretation. This mechanism was not strongly reflected in the empirical data. Within the studied R&D context, participants did not describe generative AI as a single source of truth, but rather emphasized caution regarding hallucinations and a risk of overreliance on AI-generated output. Instead of settling on AI-generated suggestions, team discussions were generally described as relying on shared human elaboration of arguments. However, in relation to R&D sensemaking the findings suggest a lighter mechanism than Ben-Shahar et al. (2025). By supporting individuals structuring and strengthening their ideas, generative AI may allow some interpretations to be used in discussion in a more polished or rhetorically convincing way than they otherwise would have. Generative AI may therefore shape collective sensemaking indirectly without being explicitly recognized as an autonomous authority within the discussion.

Overall, the findings suggest that generative AI played a more indirect role in collective sensemaking. It was mainly described as useful for preparing individual contributions, while its role in team discussions appeared weaker. Within the studied cases, consensus appeared to remain grounded in human discussion, expertise and professional experience. Generative AI was therefore not described as an autonomous authority in collective sensemaking but rather appeared to function as an individual support tool prior to team discussions. Nevertheless, by helping users to refine their arguments, generative AI may therefore indirectly shape which interpretations enter collective sensemaking in a more persuasive way.

5.4 Confidence in AI output is based in human verification and expertise

According to sensemaking theory, interpretations must build a sufficiently plausible base for action (Weick, 1995; Weick et al., 2005). When AI contributes to sensemaking, the question is therefore not only whether it provides sufficient information, but also whether actors are confident enough in AI-supported interpretations to rely on them for decisions. This issue was addressed in sub research question 3:

How does generative AI influence the practices under which R&D actors build confidence in decisions?

The empirical findings suggest that generative AI did not have a uniform influence on confidence about its use in R&D sensemaking within the studied cases. On one hand, generative AI appeared to increase confidence by broadening the information base, generating additional perspectives or structure decision material. At the same time, it appeared to also decrease participants confidence around the output because of hallucinations or overly agreeable, simplified or contextually insufficient outputs. This suggests that confidence in AI-supported sensemaking depended not only on the results themselves, but also on the interpretation of those and whether they could be verified and owned by the user. The empirical findings therefore indicate two conditions regarding the confidence in AI-supported sensemaking, including the task stakes and the extent to which outputs could be verified by a human actor with relevant expertise.

First, task stakes appeared to influence how much confidence participants were willing to put into the generative AI outputs. For lower stakes task such as early exploration, generative AI was described to be useful, even when outputs had to be treated with caution, while for higher stakes tasks such as verification or consequential decisions there appeared to be a higher threshold for confidently relying on generative AI outputs. This aligns with research form Martin and Waldmann (2023), who argue that perceived legitimacy of algorithmic decisions decreases with increased decision importance (Martin & Waldman, 2023). Even though the research is not specifically about confidence in AI-supported sensemaking, it helps understanding why generative AI outputs related to higher stakes tasks appeared to require stronger human judgement and verification.

Second, confidence appeared to depend on whether outputs could be verified by a human expert. Participants reported that they were more confident in AI contributions that had been reviewed by individuals with high domain knowledge, as those were attributed the capability to detect errors and contextualize the output. This relates to theoretical work of de Fine Licht and de Fine Licht (2020), who argue that perceived legitimacy of algorithm-supported decision-making depends less on full tool transparency, but more on explainability of the output (de Fine Licht & de Fine Licht, 2020). The empirical findings extend this argument by suggesting that in R&D sensemaking explainability does not necessarily need to originate from the generative AI system itself but can also be mediated by human interpretation and domain expertise. The findings do also relate to a concern raised by Keding and Meissner (2021) in research on AI use for R&D investment decisions. They argue that AI may lead to higher perceived decision quality, thereby introducing a risk of overreliance (Keding & Meissner, 2021). In the cases studied, participants recognized risks around hallucinations and plausible yet incorrect results, emphasizing the need for human verification of generative AI outputs. Human expertise and verification appeared not only to increase confidence levels in AI-supported interpretations, but also to serve as a perceived safeguard against overreliance. However, the interview-based methodology does not allow a conclusion about whether participants' reported caution actually prevented overreliance in practice.

The emphasis on human judgement in making generative AI output credible throughout the empirical data aligns with the automation augmentation paradox by Raisch and Krakowski (2021). They argue that AI adds value not simply by replacing human actors, but rather when automation and augmentation are combined in a suitable way (Raisch & Krakowski, 2021). The empirical findings suggest that generative AI enhanced participants' confidence for some tasks, while human judgement and responsibility remained essential in consequential contexts. In the studied cases in R&D, confidence in AI-supported interpretations and decisions appeared to be a matter of combining generative AI capabilities with the users' domain expertise and contextual understanding.

Overall, the findings suggest that confidence in AI-supported sensemaking is conditional. Even though generative AI can broaden the user's information access, generate perspectives or prepare structured material for decisions, AI tools themselves were not described to provide a sufficient basis for confident action. Rather, confidence appeared to depend on the task stakes, the possibility of verifying outputs and the expertise of the human interpreter. Therefore, although generative AI could

contribute to building confidence in some cases, credibility and ownership of the resulting interpretations and decisions remained based in human judgement.

5.5 Uncertainty is shifted rather than removed

The empirical findings suggest that generative AI did not simply reduce uncertainty in R&D sensemaking but rather shifted it into different uncertainty domains. Participants repeatedly reported that generative AI helped them to accelerate and broaden information search, ease the access into unfamiliar domains or summarize large amounts of data. This indicates that generative AI may reduce uncertainty related to missing information. At the same time, participants reported that new uncertainty emerged with the use of generative AI tools related to concerns around hallucinations, agreeability, output sensitivity to prompting and accessibility of contextual information. This suggests that the use of generative AI in R&D sensemaking did not lead to a straightforward reduction of uncertainty in the studied cases, but rather that uncertainty was redistributed.

Milliken (1987) argues that uncertainty is partially perceptual and distinguishes between uncertainty about the state of the environment, about the effect of a certain event and about the organization's available responses (Milliken, 1987). Within the scope of the empirically examined cases generative AI appeared most useful to reduce uncertainty based on lack of information, resembling Milliken's state uncertainty. It helped participants to resolve uncertainty around determining what information exists and where to find it but also by analysing and summarizing large amounts of data. The data suggests that generative AI was perceived as particularly useful in early exploration and interpretation of information. At the same time, effect and response uncertainty did not appear to be simply resolved by more information. Those types of uncertainty are more context dependent and require actors to interpret how an organization is affected and may react to certain situation. The findings indicate that generative AI could resolve these issues only conditionally, depending on whether relevant organizational or project context was supplied to the tool. Without access to such context, generative AI responses were often described as too generic, as they could not account for specific constraints and therefore did not provide a reliable basis for action. This nuances the capability of generative AI to reduce uncertainty. The reduction of uncertainty, especially around effect and response uncertainty, appeared to be strongest when relevant information was accessible or could be supplied. Across the studied cases, this possibility often appeared to be limited by confidentiality concerns, risk of data leakage and the limited integration of generative AI tools with internal systems.

Literature around AI-supported decision-making addresses a similar issue. Wu & Shang (2020) argue that AI may reduce uncertainty by processing large amounts of information and thereby support decision making under uncertainty. At the same time, they warn that AI may also introduce risks around unfamiliar situations, boundary scenarios and situations with incomplete data (Wu & Shang, 2020). The empirical findings align with this, suggesting that generative AI appeared helpful in reducing information-related uncertainty, but weaker in situations that required interpretation of company specific context, particularly when this context could not be provided to the AI model.

Lastly, Berente et al. (2021) argue that contemporary AI is autonomous, learning and inscrutable. The concept of inscrutability in particular highlights the black-box characteristic of AI (Berente et al., 2021), which may create a need for users and organizations to assess to what extent AI-generated outputs can be relied on and to determine accountability for their use. This is relevant for the empirical findings, which surface a related issue. The data suggests that, in R&D sensemaking, generative AI outputs themselves appeared to become objects of user interpretation. Participants repeatedly reported the need to be aware of limitations and highlighted the importance of human verification before relying on generative AI outputs. In terms of sensemaking, generative AI therefore did not only provide cues for interpreting the initial problem but became a cue for interpretation itself.

Overall, the findings suggest that generative AI shifted uncertainty rather than removing it. In cases where generative AI made information more accessible, users shifted from searching for information to assessing whether the AI-researched information was reliable and contextually appropriate. This indicates that the capability of generative AI to reduce uncertainty did not rely on information processing alone, but also on whether users could utilize the outputs reliably for the given R&D context.

6 Toward a framework of generative AI in R&D sensemaking

The tentative conceptual framework presented in section 2.4 framed the influence of generative AI on sensemaking primarily around sensemaking activities. However, the empirical findings did not support organising generative AI involvement in sensemaking solely around a fixed sequence of activities. As discussed in section 5.1, generative AI use in R&D sensemaking was also conditional on the task stakes, the task domain, the social sensemaking mode, the contextual embeddedness of the tool and the expertise of the human interpreter. The revised conceptual framework organizes these conditions in two dimensions: The contextual fit dimension captures the relation of task, tool and accessible context, while the accountability demand dimension captures the extent to which the outputs require verification and expert ownership. A more detailed elaboration of these dimensions follows with the subsequent section.

6.1 The axes: Framework dimensions

The **contextual fit** dimension addresses the degree to which the tool has access to relevant context and how well it fits the task. This captures the conditions task domain and contextual embeddedness. Contextual fit is high when the tool is given access to relevant, context-specific information or when the task fits the nature of the tool. The contextual fit is low when the generative AI tool lacks the relevant organizational context or when the task requires specialized knowledge that the tool cannot provide.

The **accountability demand** dimension addresses the degree of verification and human ownership required for a generative AI output to be perceived a legitimate and sufficiently reliable contribution to a given sensemaking task. The accountability demand is comparatively lower for divergent activities or lower stakes tasks. This includes activities such as exploration, initial orientation or building of provisional interpretations, where outputs still require critical assessment, but are not relied on for consequential action. The accountability demand becomes higher, when output contributes to convergence on team-level interpretations, technical results or decision justification. In these settings, generative AI output must be verified and interpreted by an accountable domain expert before it can be used as a credible contribution.

Figure 3 presents the revised framework as a 2x2 matrix and thereby distinguishes four roles of generative AI in R&D sensemaking. The resulting roles in the four quadrants are elaborated in the following section.

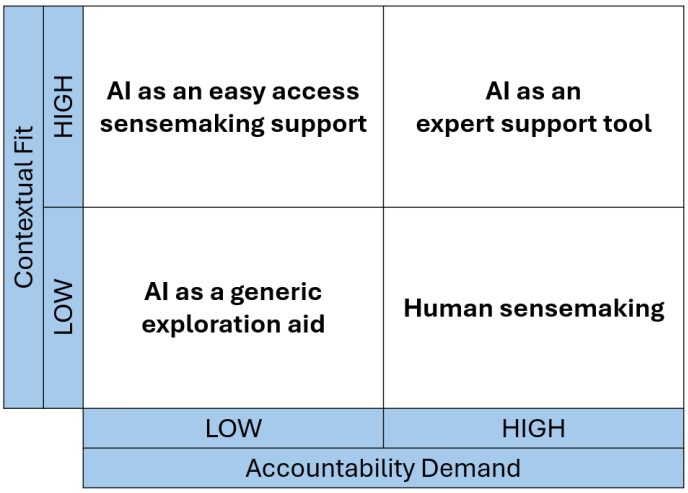


Figure 3: Fit-Accountability Framework for sensemaking for generative AI in R&D sensemaking

6.2 The four Quadrants: Roles of generative AI in sensemaking

First, when contextual fit is high and the accountability demand low, generative AI can be seen as an **easy access sensemaking support**. In this quadrant, generative AI is particularly useful in early and divergent sensemaking activities, such as exploration, new domain entry or when searching for answers in available project-specific material. The frequently described use of generative AI as a sparring partner to iterate and refine ideas also belongs in this category, when the tool is given enough contextual knowledge. One recurring pattern in this quadrant was the accessibility of generative AI, allowing it to give users quick insights to emerging questions. In these cases, it helped users develop an initial understanding before involving others. However, this accessibility did not replace the expertise of trusted colleagues, it rather offered an initial preliminary support to human sensemaking. Importantly, the lower accountability demand in this quadrant did not eliminate the need for human verification. Within the cases captured by the data, tasks appeared to be lower-stakes and easier to verify. For example, users could compare outputs directly to source material by checking referenced documents. The verification in these cases did therefore not require the same degree of accountable domain expertise as in higher-stakes tasks.

Second, when both contextual fit and accountability demand are high, generative AI functions as an **expert support tool**. In this quadrant, generative AI can be useful under certain conditions. It may help finding information, comparing interpretations or structuring arguments for the preparation of decision material. Furthermore, generative AI may support technically competent users working on coding problems with outputs reviewed and tested by the user. Importantly, the output must be interpreted and owned by an individual with a high degree of domain knowledge to be perceived legitimate as a result or as an input to a team discussion. Within the studied cases generative AI outputs were not seen as a discussion authority by itself.

Third, when both contextual fit and accountability demand are low, generative AI serves as a **generic exploration aid**. In this quadrant, it may support brainstorming, act as an advanced search engine, or generate ideas and alternative phrasing. However, as the tools lack a strong context or domain fit, its use is limited to task that allow for a more generic elaboration. In the empirical data, use cases in this quadrant therefore often treated generative AI outputs as an inspiration or a starting point that required further refinement and elaboration by a human interpreter. The user must be aware of limitations given the lack of contextual knowledge, and outputs should therefore be treated as a starting point rather than a reliable interpretation.

Fourth, when the contextual fit is low, but the accountability demand is high, **human sensemaking** remains the prevailing activity. The combination of a weak contextual fit and high accountability requirements creates a risk of generative AI misleading the user into consequential misjudgement. AI tools may still support marginal activities such as assisting formatting or phrasing, but important activities around interpretation and decision justification should remain with human experts.

The framework should not be seen as permanently categorizing a task into the same quadrant. Instead, as the purpose and consequences of generative AI support may change throughout sensemaking activities, sensemaking incidents may move across the quadrants over time. For example, generative AI may be used for generic exploration or as an easy access sensemaking support in the beginning. As the incident moves closer to a decision or a final result, the accountability demand may increase. This requires higher levels of human expert judgement and may limit the role of generative AI. The framework is based on the forms of AI use observed in this study, where generative AI did not appear as an autonomous authority in collective sensemaking. Its applicability may therefore be limited for settings where AI is treated as an accountable contributor to discussions.

7 Conclusions

This chapter concludes the thesis by presenting the main conclusions relating to the research aim and the research questions. It further presents implications for theory and practice, as well as limitations and directions of future research.

7.1 Main conclusions

This thesis examined how generative AI tools shape individual and collective sensemaking in R&D. The findings suggest that generative AI entered sensemaking conditionally. Its role depended on the type of sensemaking activity, whether the activity was individual or collective, the characteristics of the task, the user's domain expertise and whether the tool could be provided with enough contextual information. Generative AI tools were primarily described as useful in exploratory and interpretive activities, including exploration, entering new domains or using the tools as a sparring partner to refine interpretations. These activities helped users to create a starting point for further reasoning.

The effect of generative AI on interpretive plurality appeared to be less dependent on the tool itself, but more dependent on the user's interaction with the tool. Generative AI could support generating new perspectives, when actively prompted to do so. When the AI tool was instead used to elaborate on an idea input, it could reinforce this initial framing by the user and thereby lead to fixation rather than plurality.

While generative AI was frequently described to support information search and interpretation on an individual level, its role in collective sensemaking appeared to be more indirect within the studied cases. Generative AI was often used for individual preparation before team discussions, but it was not described as an autonomous contributor to collective sensemaking. Instead, participants repeatedly emphasized the importance of human interaction and expertise when discussing within the team.

The impact of generative AI on the user's confidence in AI-supported sensemaking was ambivalent. Some users reported increased confidence because of increased access to information. Others reported decreased confidence because of concerns around hallucinations, overconfident outputs and limited contextuality. Whether generative AI outputs were treated as legitimate contributions appeared to depend primarily on whether outputs could be verified and whether the verifying user had sufficient expertise to do so.

Finally, the findings suggest that generative AI shifts uncertainty rather than removing it. It may reduce uncertainty caused by missing information, but uncertainty around the implications for the specific organizational context remained. In addition, AI introduces uncertainty around output reliability and hallucinations.

7.2 Theoretical Implications

The thesis nuances theory on AI in R&D sensemaking by providing empirical insights into the role of generative AI tools. Regarding AI in R&D sensemaking, the findings contribute to literature in four ways. First, the findings suggest that generative AI was perceived more useful for exploratory and divergent activities in the sensemaking process, while convergent activities and collective decisions appeared to remain primarily human-led domains in the studied cases. Second, generative AI did not enter collective sensemaking as an autonomous actor but contributed more indirectly through individual preparation to team discussions throughout the studied cases. Third, generative AI use in R&D sensemaking did not appear to simply remove uncertainty, but to shift it away from scarcity of information toward uncertainty related to tool reliability, output verification and contextuality. Last, a framework was proposed to describe the use of generative AI in R&D sensemaking based on

contextual fit between tool and task and on the accountability demand for relying on AI output for a given task.

7.3 Practical Implications

The findings suggest several practical implications for R&D organizations looking to utilize generative AI in sensemaking. First, organizations should apply generative AI in a more task-sensitive way rather than treating adoption as a general solution. The proposed framework can support organizations in identifying and distinguishing tasks with different levels of contextual fit and accountability demand. Generative AI appears particularly useful in exploratory activities such as information search, new domain orientation or generation of new perspectives. On the other hand, more convergent activities involving higher accountability or more consequential commitments should remain human-led, with generative AI being treated more as an additional input rather than as an autonomous decision authority.

Second, organizations should recognize that human involvement in generative AI-supported sensemaking is not only a safeguard against tool error, but also a source of legitimacy. Human actors can be questioned, held accountable and embedded in existing organizational processes and social relationships, making them important in building confidence in decisions. Generative AI may allow for efficiency gains in individual preparations and generation of initial interpretations. However, the output's legitimacy is weakened if the AI contribution cannot be verified and owned by a responsible actor. Reducing human involvement is therefore only reasonable when outputs can be independently verified and accountability demand, contextual requirements and consequences of errors are minor.

Finally, organizations should treat adoption and implementation of generative AI as an effort to build capabilities rather than only offering access to a tool. Employees should be trained in prompting generative AI, but also in appropriate contestation of its output. Training should include identification of hallucinations, checking the contextual fit and evaluating outputs against credible sources and relevant organizational knowledge. Employees should also learn to use AI as a sparring partner to ask for counterarguments or alternative interpretations to broaden rather than to narrow down their own initial perspective. Such use requires appropriate conditions, including secure access to relevant organizational knowledge, clear rules for confidential information and properly maintained documentation. Without such investments in employee capability and a suitable contextual infrastructure, generative AI may still support generic exploration, but a comprehensive contribution to R&D sensemaking is likely to remain limited.

7.4 Limitations

The thesis has several methodological and empirical limitations. First, the empirical scope of 25 interviews with 18 individuals across two firms limits transferability of the findings to a broader range of R&D organizations. Further, the retrospective interview design did not observe sensemaking happening in practice. This may have introduced a bias toward successful or memorable cases of generative AI use and limited access to multiple perspectives around shared collective sensemaking episodes. Last, the study was conducted during an ongoing adoption phase, as both organizations were still integrating AI tools into their infrastructures and work processes. The findings therefore may reflect early-stage adoption practices rather than more established patterns of sensemaking supported by generative AI.

7.5 Future work

Building on the findings and limitations of this study, future research should examine whether the identified patterns of AI-supported sensemaking persist across broader organizational contexts. In

particular, the results and the proposed framework should be tested on a larger sample and through a methodological approach that observes collective sensemaking happening in real time. Future research should also include cases with higher levels of contextual embeddedness of AI tools, where generative AI may be treated as an autonomous contributor to team discussions and decisions.

The present study shows that different participants used AI tools to generate alternative perspectives or to reinforce their initial ideas, but it does not examine whether these different practices lead to different discussion outcomes. Therefore, research should examine how different forms of tool interaction affect the outcomes of individual and collective sensemaking. The role of user expertise should also be investigated more closely, as this study suggests that greater user domain expertise influences the ability to verify outputs and the perceived legitimacy of generative AI outputs in sensemaking.

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Appendix A: Semi-Structured Interview guide

Disclaimer: *The Interview guide used the broader term 'AI' instead of 'generative AI' in some questions. However, the context of the analyzed interview material primarily concerned generative AI tools, especially based in large language models.*

Overall structure

1	Opening	5 min
2	Role and decision background	10 min
3	Incident identification	10 min
4	*If suitable Incident identified:* Incident Exploration	30 min
5	General reflections on AI in sensemaking	30 min
6	Closing	5 min

Interview guide

1 Opening

1. Introduce researcher background and research purpose
2. Introduce Interview subject
3. Request permission to record interview for research purposes

2 Role and decision background (10 min)

1. Briefly describe your role and background
2. What kinds of decisions do you usually get involved in?
3. What makes you confident enough to move forward with a decision?
   Follow-ups:
   - *What type of projects have you spent the most time on recently?*
   - *Whose perspectives are usually important in decisions?*

3 Identification of relevant sensemaking incident or work situation (10 min)

Starter-Question: Can you tell me about a recent decision or situation where the path forward was not immediately obvious?

Collection of prompts used to find incident:

- **Decision under uncertainty:** Can you tell me about a recent situation where you did not feel that you had enough information to move forward?
- **Conflicting viewpoints / ambiguity:** Do you remember a situation where different people interpreted the same information differently or had different views on the right way forward?
- **Novelty:** Can you remember a recent situation where you were dealing with something new?
- **Unexpected Result:** Can you tell me about a situation where something did not behave as expected?
- **Trade-Off:** Can you share a time when you had to choose among options that ll had clear, serious downsides?
- **Decision revision:** Can you recall a situation where an initial decision, assumption or approach later had to be revised?

4 Incident exploration (30 min)

1. Could you walk me through what happened from when the situation first appeared until a decision was reached?
   Follow-Up:
   - *Who was involved?*
   - *What were the main interpretations / options?*
   - *What did it take to move forward together?*
2. Was AI involved in any part of this process?
   Follow-Up:
   - *In what part of the process / for which activities?*
3. [If AI used] Did AI introduce any new perspectives?
4. [If Ai used] Did AI affect how the team discussed the issue or reached convergence?
   Follow-Up:
   - *Why? / Why not?*
   - *Was it questioned?*
5. Did AI use have an influence on your confidence in the final decision?
   Follow-Up:
   - *Why? / Why not?*
   - *[If not] What would it take for you to make you more confident?*
6. [If no AI used] Why was AI not used?
   Follow-Up:
   - *Was AI use not considered or actively avoided?*
   - *Why? / Why not?*
7. If this process was done with / without the use of (more / less) AI tools, what would have been different?

If time left, selectively ask questions from section 5, otherwise proceed to closing

5 General reflections on AI and sensemaking (15-30 min)

1. What role does AI currently play in your everyday work?
   Follow-Up:
   - *Do you use AI individually or within the team?*
   - *Can you give an example of a situation where AI helped you with a given problem?*
   - *What are your primary usecases of AI?*
2. Are there any cautions or limitations for use of AI?
3. When using AI for interpretation / sparring, do you use it to check or to challenge your own idea?
   Follow-Up:
   - *Why?*
   - *Does AI only have impact on speed, or also on quality of your outputs?*
4. Has AI ever helped you discover information or open new perspectives that you would otherwise not have considered?
5. Does AI have an influence on your confidence in a decision / problem that was supported by AI?
   - *Why? / Why not?*
   - *[If not] What would it take for you to make you more confident?*
6. Does AI play a role in team discussions?
   Follow-Up:
   - *Do you think AI affects the way teams discuss an issue or reach convergence?*
   - *How do you treat a teammates contribution to a discussion if it was generated or supported by AI?*

7. In which situations would you ask an AI tool for feedback rather than a colleague, and in which situations would you prefer asking a colleague?
   Follow-Up:
   - *Is AIs benefit based primarily on accessibility or on output quality?*
   - *Do you treat AI more as a team member or as a tool?*

Broader reflection on sensemaking

8. Stepping back from specific cases, how do you think AI affects the way individuals make sense of uncertain situations?
9. How do you think does it affects the way teams make sense of uncertain situations?

6 Closing (5 min)

- Is there anything important regarding AI, uncertainty or sensemaking that we have not discussed?
- [If needed] Ask for potential participants

Appendix B: List of Interview material included in Analysis

List of study participants round one. Role descriptions anonymized. Durations rounded to 5 minutes

First round of Interviews		
Role	**Company**	**Approximate Duration**
Engineering Professional	Company A	60 min
Engineering Professional	Company A	60 min
R&D Manager	Company A	70 min
R&D Manager	Company A	55 min
Engineering Professional	Company B	60 min
Engineering Professional	Company B	55 min
Innovation Professional	Company B	55 min
Innovation Professional	Company B	50 min
R&D Manager	Company B	60 min
R&D Manager	Company B	60 min
R&D Manager	Company B	50 min
R&D Manager	Company B	50 min

List of study participants round two. Role descriptions anonymized. Durations rounded to 5 minutes

Second round of Interviews		
Role	**Company**	**Approximate Duration**
Engineering Professional	Company A	55 min
Engineering Professional	Company A	55 min
Engineering Professional	Company A	50 min
Engineering Professional	Company A	50 min
R&D Manager	Company A	30 min
Engineering Professional	Company B	55 min
Engineering Professional	Company B	50 min
Engineering Professional	Company B	50 min
Innovation Professional	Company B	60 min
Innovation Professional	Company B	50 min
Innovation Professional	Company B	40 min
R&D Manager	Company B	50 min
R&D Manager	Company B	40 min

Appendix C1: Codebook round one

Codebook for the first round of coding
Uncertainty context / ambiguity
Cue noticing
Interpretation
Negotiation
Decision
AI involvement
Information search space
Convergence
Plurality
Debate and Contestation
Decision confidence

Appendix C2: Codebook round two

Codebook for the second round of coding
Social Mode of sensemaking
Team sensemaking
Individual sensemaking
Task characteristics
Task stakes shape AI use
Domain dependent utility of AI
Task complexity shapes AI use
Early Sensemaking / Information Search
AI use for low-stake research
AI accelerates information search
AI broadens information search
AI offers a starting point
AI as strong support in early exploration
AI accelerates new domain entry
Interpretation
AI as low-effort / low-friction substitute for colleagues
AI-supported decision preparation
AI as a sparring partner
AI lacks tacit and organizational knowledge
AI expands plurality of perspectives
AI as articulation support
AI-supported problem interpretation
AI for technical troubleshooting
Negotiation / Contestation
AI as personal support rather than team authority
Human discussion remains essential
Perspective plurality across functions
AI as weak evidence in discussion
Human credibility superior over AI
AI supports crossfunctional participation
AI too agreeable

Legitimacy and confidence
AI output sensitive to prompting
Confidentiality as limitation
Speed-confidence tradeoff
Team discussion builds confidence
Human verification of AI output
Risk of overreliance
AI confirmation increases confidence
AI decreases decision confidence
Decision / Task ownership remains human

Appendix D1: Coding Excerpt 1



Appendix D2: Coding Excerpt 2



Appendix D3: Coding Excerpt 3



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